

COSE471: Data Science

Pattern Mining

Department of Computer Science and Engineering

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Course Schedule

Week	Day	Content	HW/Project	Note
1	Mar 06	Introduction		
2	Mar 13	Data		
3	Mar 20	Pattern mining		
4	Mar 27	Tutorial 1	● HW1	<i>Latex, Preprocessing/Pattern mining</i>
5	Apr 03	Clustering	●	
6	Apr 10	Outlier Detection & Invited Talk	●	<i>An invited talk from Meta</i>
7	Apr 17	Tutorial 2	● HW2	<i>Clustering, Outlier detection</i>
8	Apr 24	Classification	●	
9	May 01	Classification	●	
10	May 08	Tutorial 3	●	<i>Classification, Feature engineering</i>
11	May 15	Mid-term exam		<i>Offline, Closed-book, No electronic devices</i>
12	May 22	Review & Project Notice	● Project	
13	May 29	Project discussion	●	
14	Jun 05	Project discussion	●	
15	Jun 12	Final presentation	●	
16	Jun 19	Wrap-up and Final report	●	

RECAP

❑ Diversity of data

- Data structure, type, state

❑ Understanding data

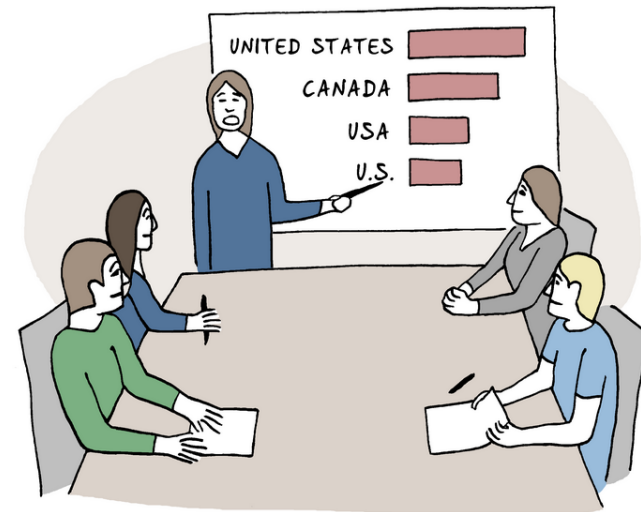
- Data sets, objects, attributes, matrix
- Data statistics
- Data visualization
- Data similarity

❑ Preprocessing data

- Data quality measures
- Data cleaning
- Data integration
- Data transformation

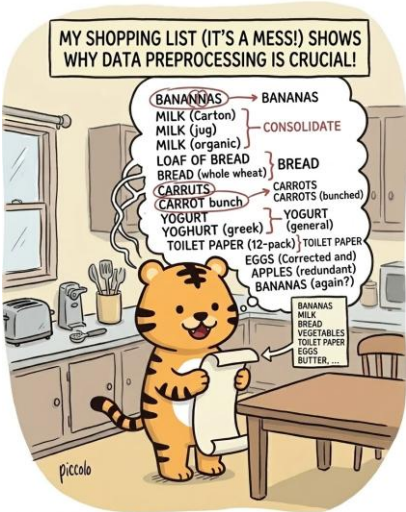


I FOUND THE BUG, BOSS. I WAS ADDING 'ID' COLUMN INSTEAD OF 'AMOUNT'.



AS YOU CAN SEE, OUR TOP MARKETS ARE UNITED STATES, CANADA, USA AND THE U.S.

RECAP



Outline

❑ Basic concepts of pattern mining

- Pattern discovery
- Frequent itemsets
- Association rule
- Closed and maximal patterns

❑ Pattern mining methodologies

❑ Pattern mining evaluation



Pattern Mining > Pattern Discovery

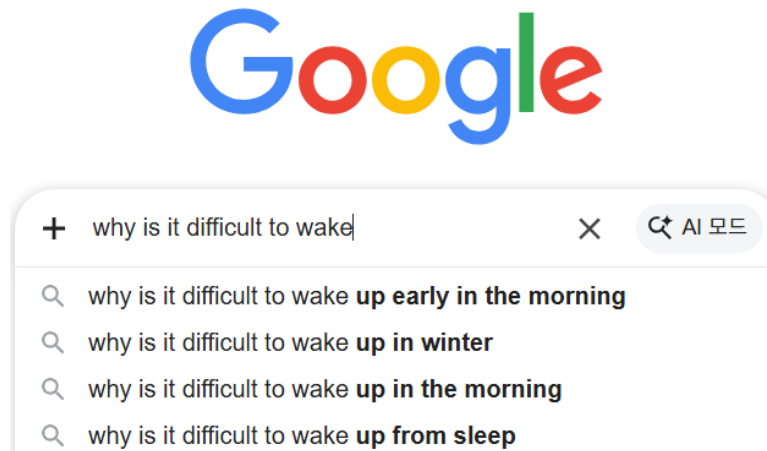
□ Pattern

- A set of items, subsequences, or substructures that occur frequently together in a data set
- Patterns represent intrinsic and important properties of data sets

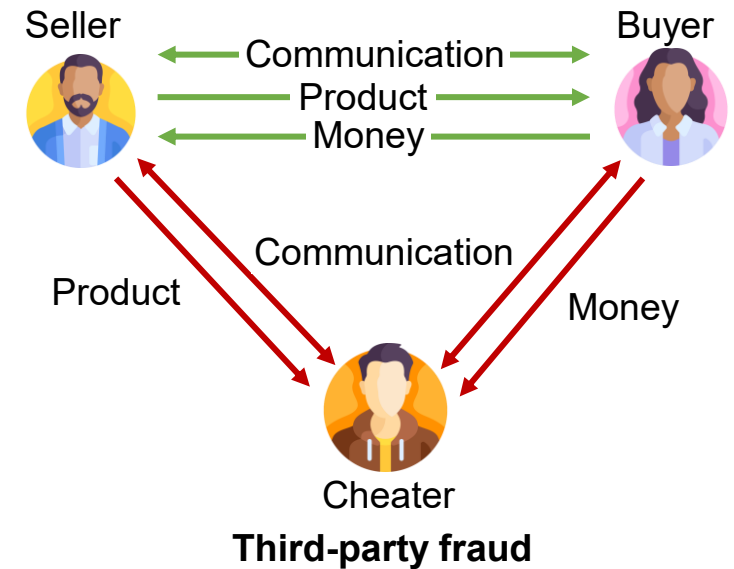
Frequent itemsets



Frequent sequences



Frequent structures



■ Pattern Mining > Pattern Discovery

□ Pattern discovery

- Uncovering patterns from massive data sets
- It can answer questions such as:
 - What products were often purchased together?
 - What are the subsequent purchases after buying an iPad?
 - What are the frequent communication patterns in social network services?

□ Foundation for many essential data science tasks

- Association, correlation, and causality analysis
- Discriminative pattern-based classification
- Pattern-based subspace clustering

Pattern Mining > Pattern Discovery

❑ Broad applications

- Market basket analysis, cross-marketing, catalog design, sale campaign analysis, Web log analysis, biological sequence analysis
- Many types of data: transactions, graph, spatiotemporal, time-series, and stream data



Pattern Mining > Frequent Itemsets

□ Transactional Database (TDB)

- Each transaction is associated with an identifier, called a TID
- May also have counts associated with each item sold

TID	Items bought together
1	Beer, Nuts, Diaper
2	Beer, Coffee, Diaper
3	Beer, Diaper, Eggs
4	Nuts, Eggs, Milk
5	Nuts, Coffee, Diaper, Eggs, Milk

Pattern Mining > Frequent Itemsets

□ Itemsets X: a set of items

▪ k-itemset: an itemset containing k items

- $X = \{x_1, x_2, \dots, x_k\}$

- e.g., X = {Beer} is a 1-itemset and X = {Beer, Nuts, Diaper} is a 3-itemset

TID	Items bought together
1	Beer, Nuts, Diaper
2	Beer, Coffee, Diaper
3	Beer, Diaper, Eggs
4	Nuts, Eggs, Milk
5	Nuts, Coffee, Diaper, Eggs, Milk

Pattern Mining > Frequent Itemsets

□ Support $s(X)$ of an itemset X

- Absolute support $\rightarrow$ count
 - The occurrences of an itemset in all transactions
 - e.g., $s(\{\text{Beer}\}) = 3$, $s(\{\text{Diaper}\}) = 4$, $s(\{\text{Beer}, \text{Eggs}\}) = 1$
 - Can be also written as $\{\text{Beer}\}:3$ for simplicity
- Relative support $\rightarrow$ ratio
 - The fraction of transactions that contain X
 - i.e., the probability that a transaction contains X
 - e.g., $s(\{\text{Beer}\}) = 3/5 = 60\%$, $s(\{\text{Diaper}\}) = 4/5 = 80\%$, $s(\{\text{Beer}, \text{Eggs}\}) = 1/5 = 20\%$
 - Can be also written as $\{\text{Beer}\}:60\%$ or $\{\text{Beer}\}:0.6$
- They can be used interchangeably
 - e.g., given 10 transactions, the absolute support for an itemset is 3 $\rightarrow$ the relative support $30\% = 0.3$

TID	Items bought together
1	Beer, Nuts, Diaper
2	Beer, Coffee, Diaper
3	Beer, Diaper, Eggs
4	Nuts, Eggs, Milk
5	Nuts, Coffee, Diaper, Eggs, Milk

Pattern Mining > Frequent Itemsets

□ Frequent itemsets

- An itemset X is frequent if $s(X)$ is no less than a minsup (minimum support threshold)
 - i.e., $s(X) \geq \text{minsup}$
- e.g., given 5 transactions and a minsup = 50% ($= 5 * 0.5 = 2.5$)
 - Frequent 1-itemsets
 - e.g., Beer: 3/5 (= 60%)
 - What about Milk?
 - Frequent 2-itemsets
 - e.g., {Beer, Diaper}: 3/5 (= 60%)
 - What about {Beer, Nuts}?
 - Frequent 3-itemsets
 - Is there any?

TID	Items bought together
1	Beer, Nuts, Diaper
2	Beer, Coffee, Diaper
3	Beer, Diaper, Eggs
4	Nuts, Eggs, Milk
5	Nuts, Coffee, Diaper, Eggs, Milk

■ Pattern Mining > Frequent Itemsets

□ Example: find ALL the frequent k-itemsets

Minsup = 40%

TID	Items bought together
1	Beer, Diaper
2	Beer, Coffee
3	Beer, Diaper, Eggs
4	Eggs
5	Coffee, Diaper, Eggs

■ Pattern Mining > Frequent Itemsets

□ Example: find ALL the frequent k-itemsets

Minsup = 40%

TID	Items bought together
1	Beer, Diaper
2	Beer, Coffee
3	Beer, Diaper, Eggs
4	Eggs
5	Coffee, Diaper, Eggs

Frequent 1-itemsets:

{Beer}: 3/5 (= 60%); {Diaper}: 3/5 (= 60%); {Eggs}: 3/5 (= 60%); {Coffee}: 2/5 (=40%)

Frequent 2-itemsets:

{Beer, Diaper}: 2/5 (= 40%), {Beer, Coffee}: 1/5 (= 20%), {Beer, Eggs}: 1/5 (= 20%), {Diaper, Coffee}: 1/5 (= 20%), {Diaper, Eggs}: 2/5 (= 40%), {Coffee, Eggs}: 1/5 (= 20%)

Frequent 3-itemsets:

{Beer, Diaper, Coffee}: 0/5 (= 0%), {Beer, Diaper, Eggs}: 1/5 (= 20%), {Beer, Coffee, Eggs}: 0/5 (= 0%), {Diaper, Coffee, Eggs}: 1/5 (= 20%),

Are they a complete set of frequent patterns? Are they interesting?

Pattern Mining > Association Rule

Minsup = 50%

TID	Items bought together
1	<i>Beer</i> , Nuts, <i>Diaper</i>
2	<i>Beer</i> , Coffee, <i>Diaper</i>
3	<i>Beer</i> , <i>Diaper</i> , Eggs
4	Nuts, Eggs, Milk
5	Nuts, Coffee, <i>Diaper</i> , Eggs, Milk

Frequent k-itemsets given a minsup = 50%

Beer: 3/5 (= 60%); Nuts: 3/5 (= 60%);
Diaper: 4/5 (= 80%); Eggs: 3/5 (= 60%);
{*Beer*, *Diaper*}: 3/5 (= 60%)

□ Association rule is more telling than just frequent itemsets

- e.g., for the frequent 2-itemset {Diaper, Beer}, we may expect
 - *Buying diapers may likely lead to buying beers*
 - *Buying beers may likely lead to buying diapers*
- } Which one is more appealing?

Pattern Mining > Association Rule

□ Association rule

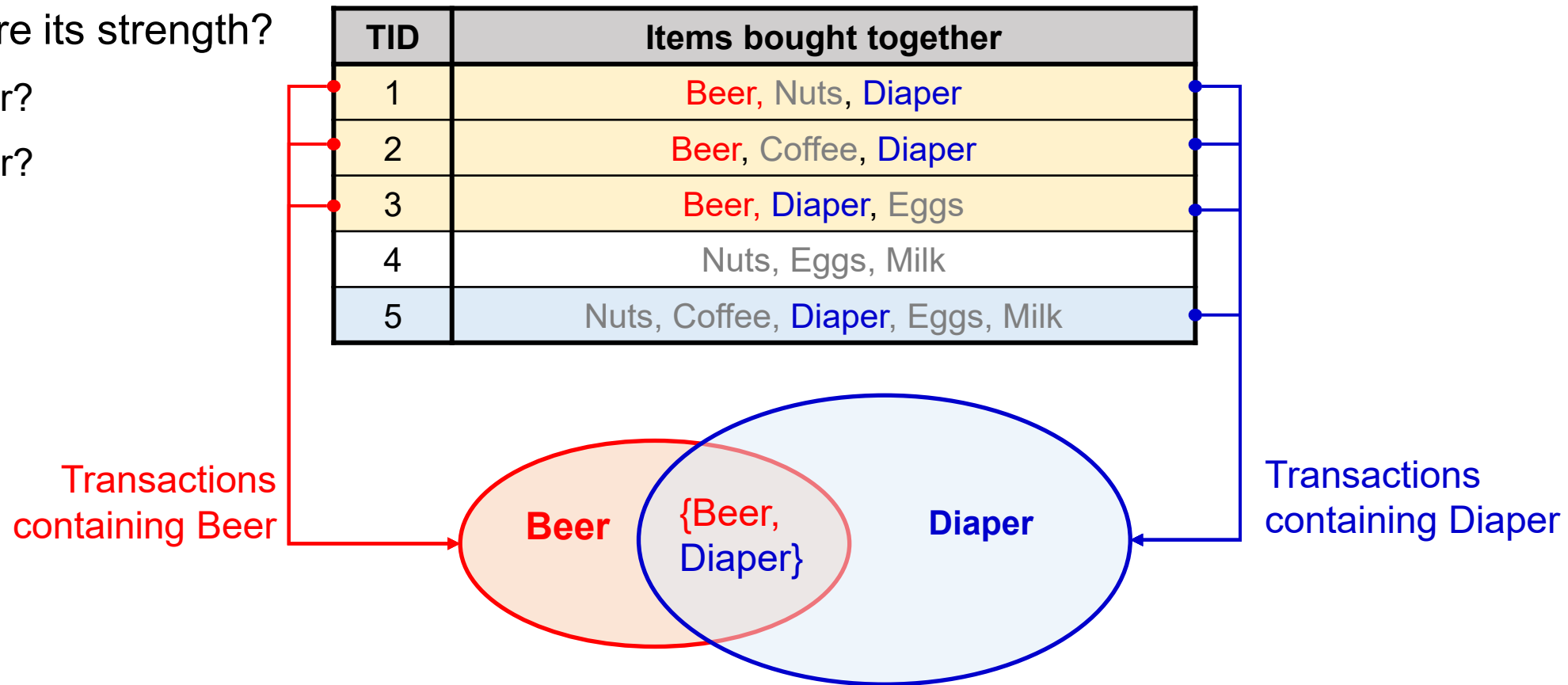
- Given itemsets X and Y, $X \rightarrow Y$
 - i.e., if X, then Y: when X occurs, then Y will as well
- How to measure its strength?
 - Beer $\rightarrow$ Diaper?
 - Diaper $\rightarrow$ Beer?

TID	Items bought together
1	Beer, Nuts, Diaper
2	Beer, Coffee, Diaper
3	Beer, Diaper, Eggs
4	Nuts, Eggs, Milk
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Pattern Mining > Association Rule

□ Association rule

- Given itemsets X and Y, $X \rightarrow Y$
 - i.e., if X, then Y: when X occurs, then Y will as well
- How to measure its strength?
 - Beer $\rightarrow$ Diaper?
 - Diaper $\rightarrow$ Beer?



Pattern Mining > Association Rule

□ Association rule

- Given itemsets X and Y, $X \rightarrow Y$
 - i.e., if X, then Y: when X occurs, then Y will as well

□ The strength of an association rule

- Support of $X \cup Y$ (i.e., $s(X \cup Y)$)
 - e.g., $s(\{\text{Diaper}, \text{Beer}\}) = 3/5 = 60\%$
- Confidence of $X \rightarrow Y$ (i.e., $c(X \rightarrow Y)$)
 - The conditional probability that a transaction containing X also contains Y

– $c(X \rightarrow Y) = \frac{s(X \cup Y)}{s(X)}$

- e.g., the confidence of a rule Diaper $\rightarrow$ Beer?

– $c(\text{Diaper} \rightarrow \text{Beer}) = \frac{s(\{\text{Diaper}, \text{Beer}\})}{s(\{\text{Diaper}\})} = \frac{3}{4} = 0.75$

- e.g., the confidence of a rule Beer $\rightarrow$ Diaper?

– $c(\text{Beer} \rightarrow \text{Diaper}) = \frac{s(\{\text{Diaper}, \text{Beer}\})}{s(\{\text{Beer}\})} = \frac{3}{3} = 1.0$

TID	Items bought together
1	Beer, Nuts, Diaper
2	Beer, Coffee, Diaper
3	Beer, Diaper, Eggs
4	Nuts, Eggs, Milk
5	Nuts, Coffee, Diaper, Eggs, Milk

Pattern Mining > Association Rule

❑ RECAP: frequent itemsets and association rule

- An itemset X is **frequent** if $s(X)$ is no less than minsup (minimum support)
 - i.e., $s(X) \geq \text{minsup}$
- An association rule $X \rightarrow Y$ is **confident** if $c(X \rightarrow Y)$ is no less than minconf (minimum confidence)
 - i.e., $c(X \rightarrow Y) \geq \text{minconf}$
- e.g., given a minsup = 50% and minconf = 80%
 - Frequent itemsets
 - Beer: 3/5 (= 60%); Nuts: 3/5 (= 60%); Diaper: 4/5 (= 80%); Eggs: 3/5 (= 60%), {Beer, Diaper}: 3/5 (= 60%)
 - Confident (strong) association rules
 - Beer $\rightarrow$ Diaper: $c(\text{Beer} \rightarrow \text{Diaper}) = 1$ (= 100%)
 - ~~Diaper $\rightarrow$ Beer: $c(\text{Diaper} \rightarrow \text{Beer}) = 0.75$ (= 75%)~~

TID	Items bought together
1	Beer, Nuts, Diaper
2	Beer, Coffee, Diaper
3	Beer, Diaper, Eggs
4	Nuts, Eggs, Milk
5	Nuts, Coffee, Diaper, Eggs, Milk

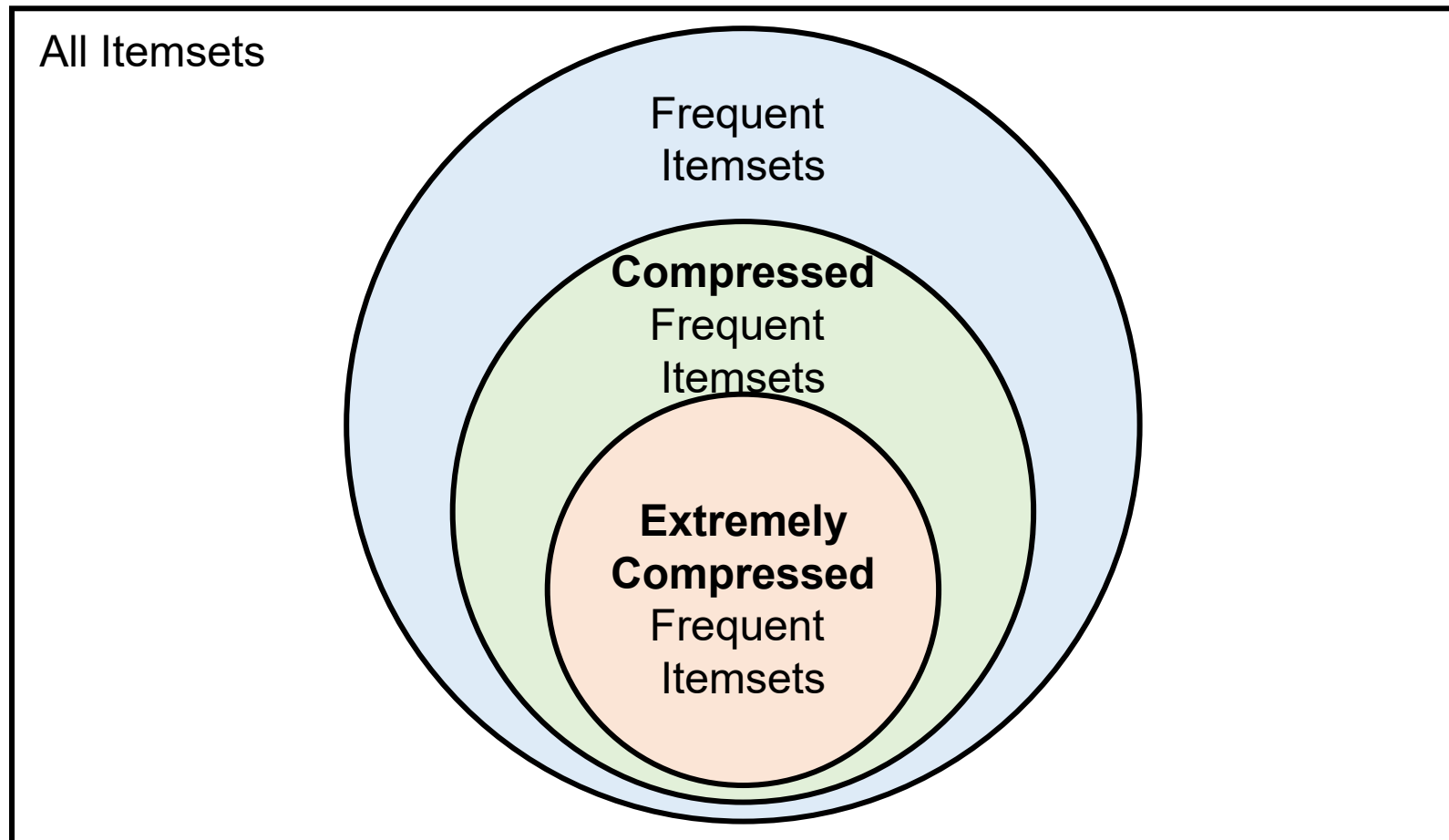
❑ Mining frequent itemsets and association rules are very close problems

- Can we think of scalable methods for large datasets?
- Can we think of minimum but still sufficient information for identifying them?

Pattern Mining > Closed and Maximal Patterns

□ Motivation

- How can we compress patterns (frequent itemsets) without losing their support information?



Pattern Mining > Closed and Maximal Patterns

❑ Issue: a long pattern contains a combinatorial number of sub-patterns

TID	Items bought together
1	$a_1, a_2, \dots, a_{49}, \mathbf{a_{50}}$
2	$a_1, a_2, \dots, a_{99}, \mathbf{a_{100}}$

- Given minsup=1, find all frequent itemsets
 - 1-itemsets:
 - 2-itemsets:
 - ...
 - 50-itemsets:
 - ...
 - 99-itemsets:
 - 100-itemset:

Pattern Mining > Closed and Maximal Patterns

□ Issue: a long pattern contains a combinatorial number of sub-patterns

TID	Items bought together
1	$a_1, a_2, \dots, a_{49}, \mathbf{a_{50}}$
2	$a_1, a_2, \dots, a_{99}, \mathbf{a_{100}}$

- Given minsup=1, frequent itemsets and their supports (i.e., $s(X) = n \rightarrow X:n$)
 - 1-itemsets: $\{a_1\}:2, \{a_2\}:2, \dots, \{a_{50}\}:2, \{a_{51}\}:1, \dots, \{a_{100}\}:1,$
 - 2-itemsets: $\{a_1, a_2\}:2, \dots, \{a_1, a_{50}\}:2, \{a_1, a_{51}\}:1 \dots, \{a_{99}, a_{100}\}:1,$
...
 - 50-itemsets: $\{a_1, a_2, \dots, a_{49}, a_{50}\}:2, \dots, \{a_{51}, a_{52}, \dots, a_{99}, a_{100}\}:1$
...
 - 99-itemsets: $\{a_1, a_2, \dots, a_{99}\}:1, \dots, \{a_2, a_3, \dots, a_{100}\}:1$
 - 100-itemset: $\{a_1, a_2, \dots, a_{100}\}:1$

Pattern Mining > Closed and Maximal Patterns

□ Issue: a long pattern contains a combinatorial number of sub-patterns

TID	Items bought together
1	$a_1, a_2, \dots, a_{49}, a_{50}$
2	$a_1, a_2, \dots, a_{99}, a_{100}$

▪ Given minsup=1, frequent itemsets and their supports (i.e., $s(X) = n \rightarrow X:n$)

- 1-itemsets: $\{a_1\}:2, \{a_2\}:2, \dots, \{a_{50}\}:2, \{a_{51}\}:1, \dots, \{a_{100}\}:1,$
- 2-itemsets: $\{a_1, a_2\}:2, \dots, \{a_1, a_{50}\}:2, \{a_1, a_{51}\}:1 \dots, \{a_{99}, a_{100}\}:1,$
...
- 50-itemsets: $\{a_1, a_2, \dots, a_{49}, a_{50}\}:2, \dots, \{a_{51}, a_{52}, \dots, a_{99}, a_{100}\}:1$
...
- 99-itemsets: $\{a_1, a_2, \dots, a_{99}\}:1, \dots, \{a_2, a_3, \dots, a_{100}\}:1$
- 100-itemset: $\{a_1, a_2, \dots, a_{100}\}:1$

$$\begin{aligned} & \left(\binom{100}{1} + \binom{100}{2} + \binom{100}{3} \right) \\ & \dots + \binom{100}{99} + \binom{100}{100} \\ & = 2^{100} - 1 \\ & = 1.26 \times 10^{30} \text{ (nonillion)} \end{aligned}$$

▪ Do we have to store all the frequent itemsets and their supports?

Pattern Mining > Closed and Maximal Patterns

□ Issue: a long pattern contains a combinatorial number of sub-patterns

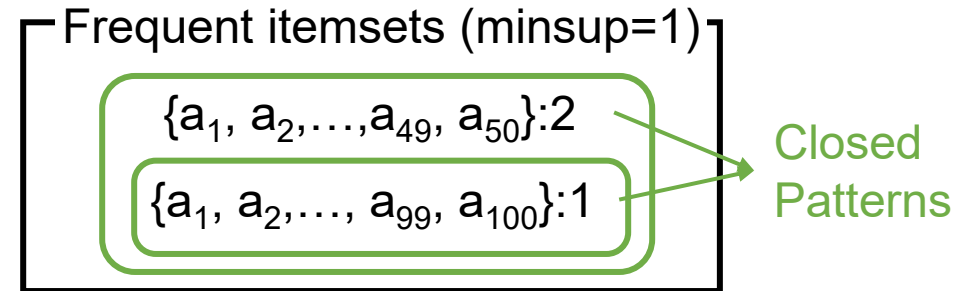
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1	$a_1, a_2, \dots, a_{49}, \mathbf{a_{50}}$
2	$a_1, a_2, \dots, a_{99}, \mathbf{a_{100}}$

- Given minsup=1, frequent itemsets and their supports (i.e., $s(X) = n \rightarrow X:n$)
 - 1-itemsets: $\{a_1\}:2, \{a_2\}:2, \dots, \{a_{50}\}:2, \{a_{51}\}:1, \dots, \{a_{100}\}:1,$
 - 2-itemsets: $\{a_1, a_2\}:2, \dots, \{a_1, a_{50}\}:2, \{a_1, a_{51}\}:1 \dots, \{a_{99}, a_{100}\}:1,$
...
 - 50-itemsets: $\{\mathbf{a_1}, \mathbf{a_2}, \dots, \mathbf{a_{49}}, \mathbf{a_{50}}\}:2, \dots, \{a_{51}, a_{52}, \dots, a_{99}, a_{100}\}:1$
...
 - 99-itemsets: $\{a_1, a_2, \dots, a_{99}\}:1, \dots, \{a_2, a_3, \dots, a_{100}\}:1$
 - 100-itemset: $\{\mathbf{a_1}, \mathbf{a_2}, \dots, \mathbf{a_{100}}\}:1$
- Are they sufficient to restore (i.e., derive) all the frequent itemsets?

Pattern Mining > Closed and Maximal Patterns

□ Expressing patterns in compressed forms

TID	Items bought together
1	$a_1, a_2, \dots, a_{49}, a_{50}$
2	$a_1, a_2, \dots, a_{99}, a_{100}$



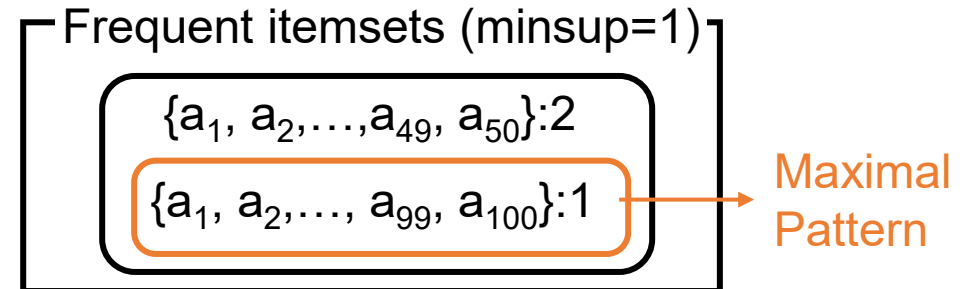
■ Closed patterns

- A frequent pattern (itemset) X is **closed** if X is frequent (i.e., $s(X) \geq \text{minsup}$), and there exists no super-itemset with the **same** support (i.e., for all $Y \supset X$, $s(Y) \neq s(X)$ or $s(Y) < s(X)$)
 - e.g., if $\text{minsup} = 1$, closed patterns will be “ $\{a_1, \dots, a_{50}\}: 2$ ” and “ $\{a_1, \dots, a_{100}\}: 1$ ”
- Closed patterns are **lossless compression** of frequent patterns
 - Reduce the number of patterns **without losing support information**
 - i.e., it is possible to exactly know the support of any frequent patterns in the closed patterns
 - e.g., $\{a_2, \dots, a_{49}\}: 2$, $\{a_5, a_{99}\}: 1$, ...

Pattern Mining > Closed and Maximal Patterns

□ Expressing patterns in compressed forms

TID	Items bought together
1	$a_1, a_2, \dots, a_{49}, a_{50}$
2	$a_1, a_2, \dots, a_{99}, a_{100}$

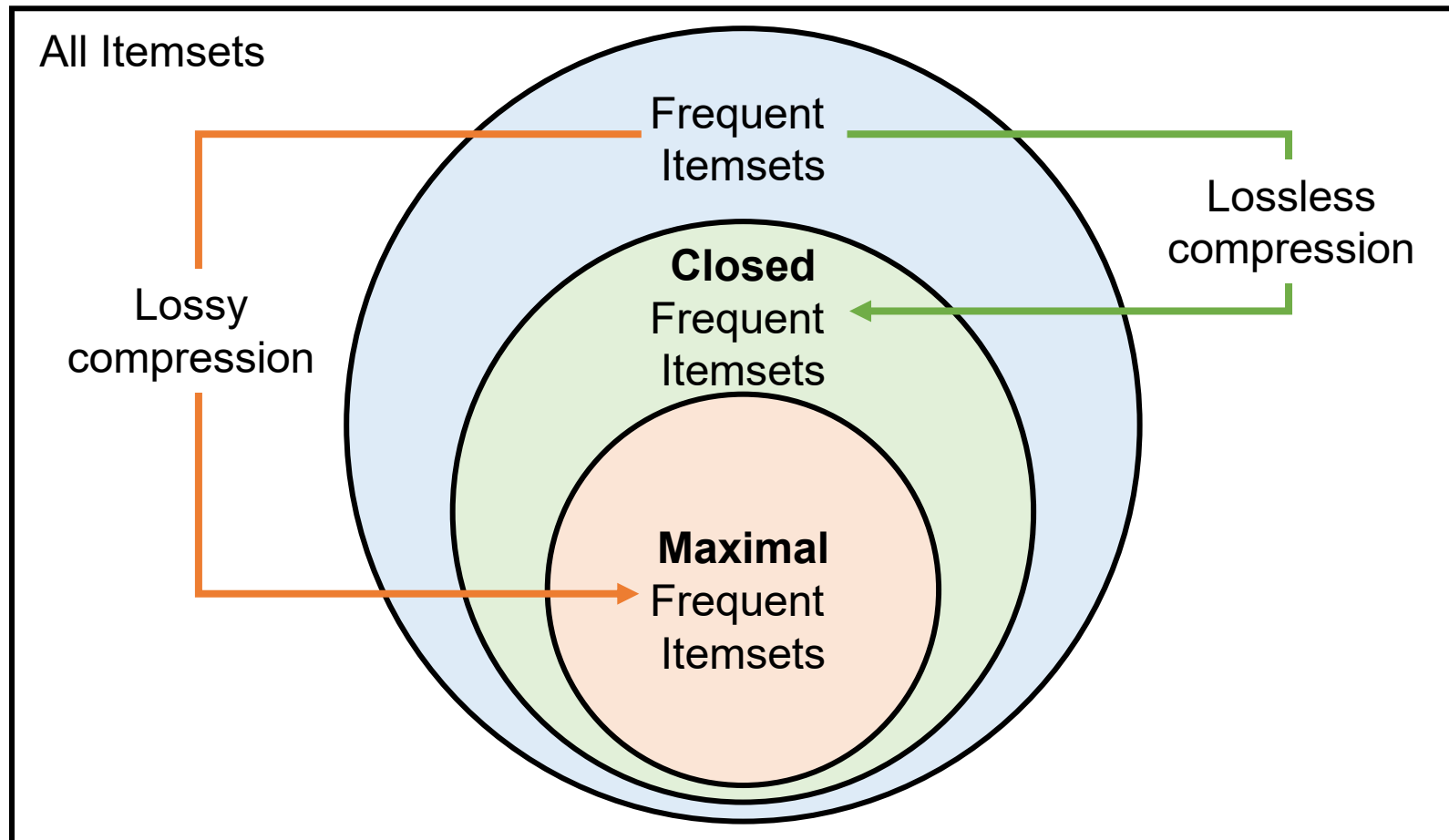


■ Maximal patterns

- A frequent pattern (itemset) X is **maximal** if X is frequent (i.e., $s(X) \geq \text{minsup}$), and there exists no **frequent** super-itemset $Y \supset X$ (i.e., for all $Y \supset X$, $s(Y) < \text{minsup}$)
 - e.g., if $\text{minsup} = 1$, maximal patterns will be " $\{a_1, \dots, a_{100}\}: 1$ "
 - What about " $\{a_1, \dots, a_{50}\}: 2$ "? There exists a frequent super-itemset " $\{a_1, \dots, a_{100}\} \supset \{a_1, \dots, a_{50}\}$ "
- Maximal patterns are **lossy compression** of frequent patterns
 - Reduce the number of patterns but **lose support information**
 - i.e., we do not exactly know the support of any frequent patterns in the maximal patterns
 - e.g., We know that $\{a_1, \dots, a_{40}\}$ is frequent, but do not know its support (which is 2)

Pattern Mining > Closed and Maximal Patterns

- Finding closed-patterns is more desirable than maximal-patterns
 - We can compress patterns without losing support information



Frequent, Closed, Maximal Patterns (Itemsets)

❑ k-itemset X

- An itemset including k items
- k can be from one to the number of unique items

❑ Frequent itemsets FX

- k-itemsets whose supports are no less than the minsup threshold
- $s(X) \geq \text{minsup}$

❑ Closed frequent itemsets CX

- Frequent itemsets with no super-itemset of the same support
- $s(CX) \geq \text{minsup}$ & for all $Y \supset CX$, $s(Y) \neq s(CX)$
- *Adding any other item to a closed frequent itemset changes its support*

❑ Maximal frequent itemsets MX

- Frequent itemsets with no frequent super-itemset
- $s(MX) \geq \text{minsup}$ & for all $Y \supset MX$, $s(Y) < \text{minsup}$
- *Adding any other item to a maximal frequent itemset makes it infrequent*

Minsup = 1

TID	Items bought together
1	Beer, Diaper
2	Beer, Coffee
3	Beer, Diaper, Eggs
4	Eggs
5	Coffee, Diaper, Eggs

All itemsets

{Beer}:3, {Diaper}:3, {Eggs}:3, {Coffee}:2,
{Beer, Diaper}:2, {Beer, Coffee}:1, {Beer, Eggs}:1,
{Diaper, Coffee}:1, {Diaper, Eggs}:2, {Coffee, Eggs}:1,
{Beer, Diaper, Coffee}:0, {Beer, Diaper, Eggs}:1,
{Beer, Coffee, Eggs}:0, {Diaper, Coffee, Eggs}:1
{Beer, Diaper, Eggs, Coffee}:0

Frequent itemsets

{Beer}:3, {Diaper}:3, {Eggs}:3, {Coffee}:2,
{Beer, Diaper}:2, {Beer, Coffee}:1, {Beer, Eggs}:1,
{Diaper, Coffee}:1, {Diaper, Eggs}:2, {Coffee, Eggs}:1,
{Beer, Diaper, Eggs}:1, {Diaper, Coffee, Eggs}:1

Closed frequent itemsets

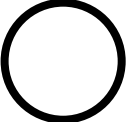
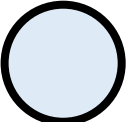
{Beer}:3, {Diaper}:3, {Eggs}:3, {Coffee}:2,
{Beer, Diaper}:2, {Beer, Coffee}:1, {Diaper, Eggs}:2
{Beer, Diaper, Eggs}:1, {Diaper, Coffee, Eggs}:1

Maximal frequent itemsets

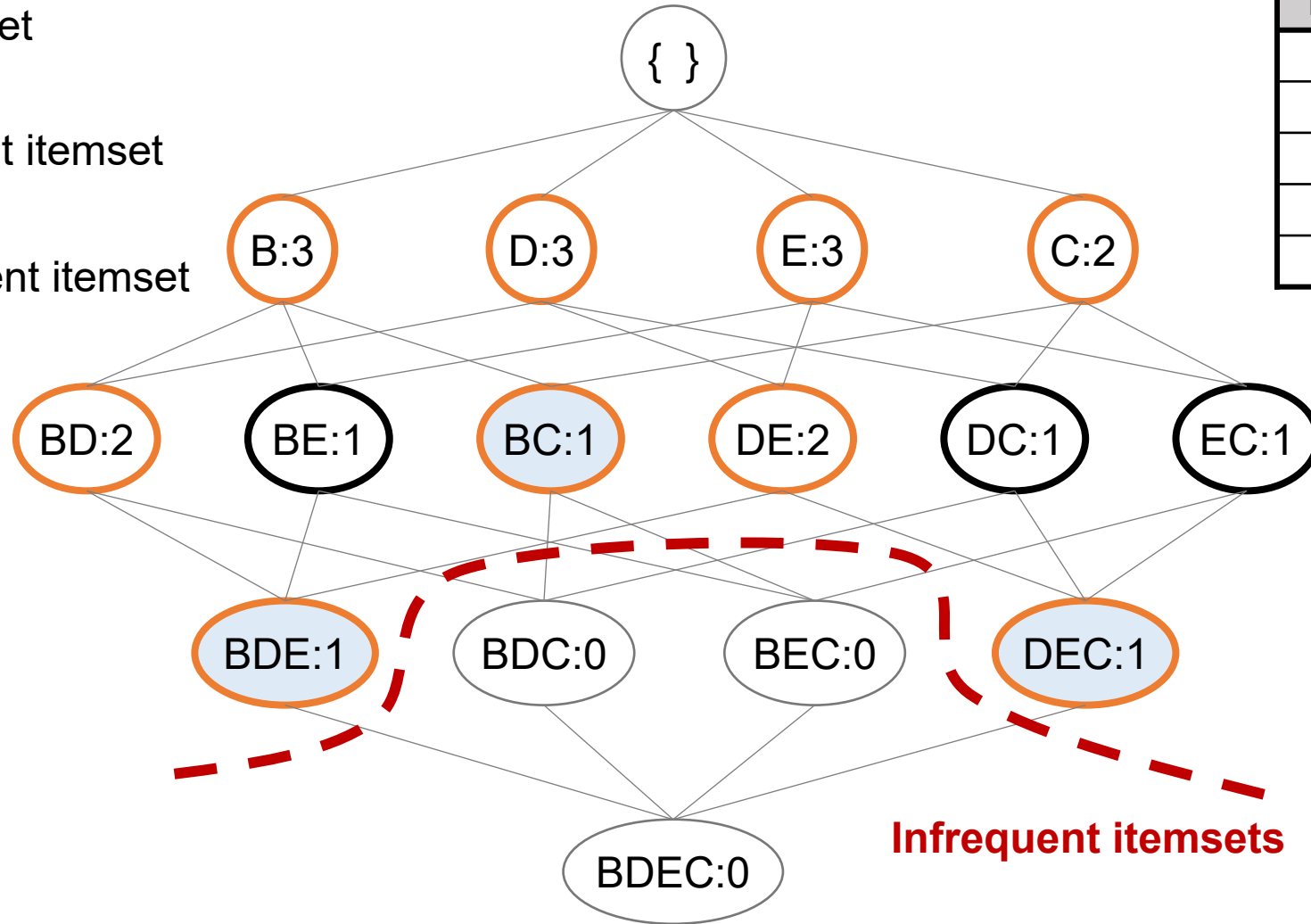
{Beer, Coffee}:1,
{Beer, Diaper, Eggs}:1, {Diaper, Coffee, Eggs}:1

Frequent, Closed, Maximal Patterns (Itemsets)

Minsup = 1

-  Frequent itemset
-  Closed frequent itemset
-  Maximal frequent itemset

TID	Items bought together
1	Beer, Diaper
2	Beer, Coffee
3	Beer, Diaper, Eggs
4	Eggs
5	Coffee, Diaper, Eggs



Infrequent itemsets

Outline

- ❑ Basic concepts of pattern mining
- ❑ **Pattern mining methodologies**
 - **Association rule discovery**
 - Apriori
 - FP-Growth
 - Rule generation
- ❑ Pattern mining evaluation

■ Pattern Mining > Methodologies > Rule Discovery

□ Given a set of transactions, we want to find all rules $X \rightarrow Y$ where

- Support $s(X \cup Y) \geq \text{minsup threshold}$
- Confidence $c(X \rightarrow Y) = s(X \cup Y) / s(X) \geq \text{minconf threshold}$

□ Example

- Association rule: Beer $\rightarrow$ Diaper ?

minsup = 50% and minconf = 80%

TDB	TID	Items bought together
	1	Beer, Nuts, Diaper
	2	Beer, Coffee, Diaper
	3	Beer, Diaper, Eggs
	4	Nuts, Eggs, Milk
	5	Nuts, Coffee, Diaper, Eggs, Milk

$$c(\text{Beer} \rightarrow \text{Diaper}) = \frac{s(\{\text{Diaper, Beer}\})}{s(\{\text{Beer}\})} = \frac{3}{3} = 100\%$$

$\nearrow = \frac{3}{5} = 60\%$
 $\searrow = \frac{3}{5} = 60\%$

We can say that Beer $\rightarrow$ Diaper is a confident rule

How to find ALL confident rules?

■ Pattern Mining > Methodologies > Rule Discovery

□ Brute-force approach 1

■ Procedure

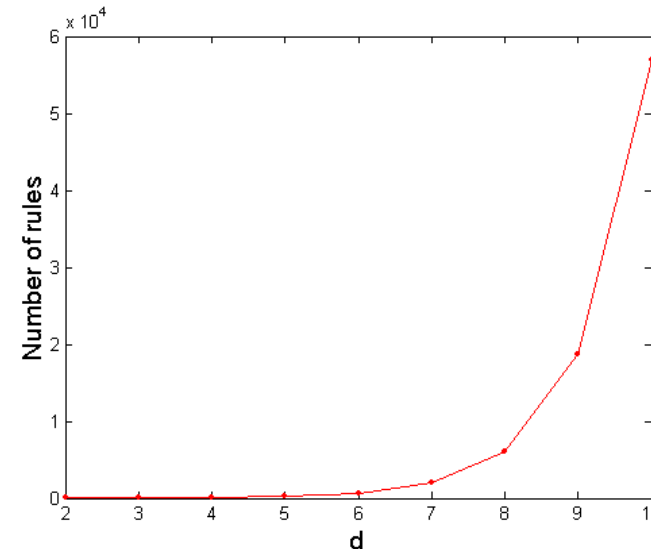
- Step 1: From the unique items, list all possible association rules
- Step 2: Compute the support and confidence for each rule
- Step 3: Prune rules with the minsup and minconf thresholds

■ Issue: computationally prohibitive!

- Given d unique items, the number of rules

$$R = \sum_{k=1}^{d-1} \left[\binom{d}{k} \times \sum_{j=1}^{d-k} \binom{d-k}{j} \right] = 3^d - 2^{d+1} + 1$$

For $d = 6$ items, $R = 602$ possible rules



Pattern Mining > Methodologies > Rule Discovery

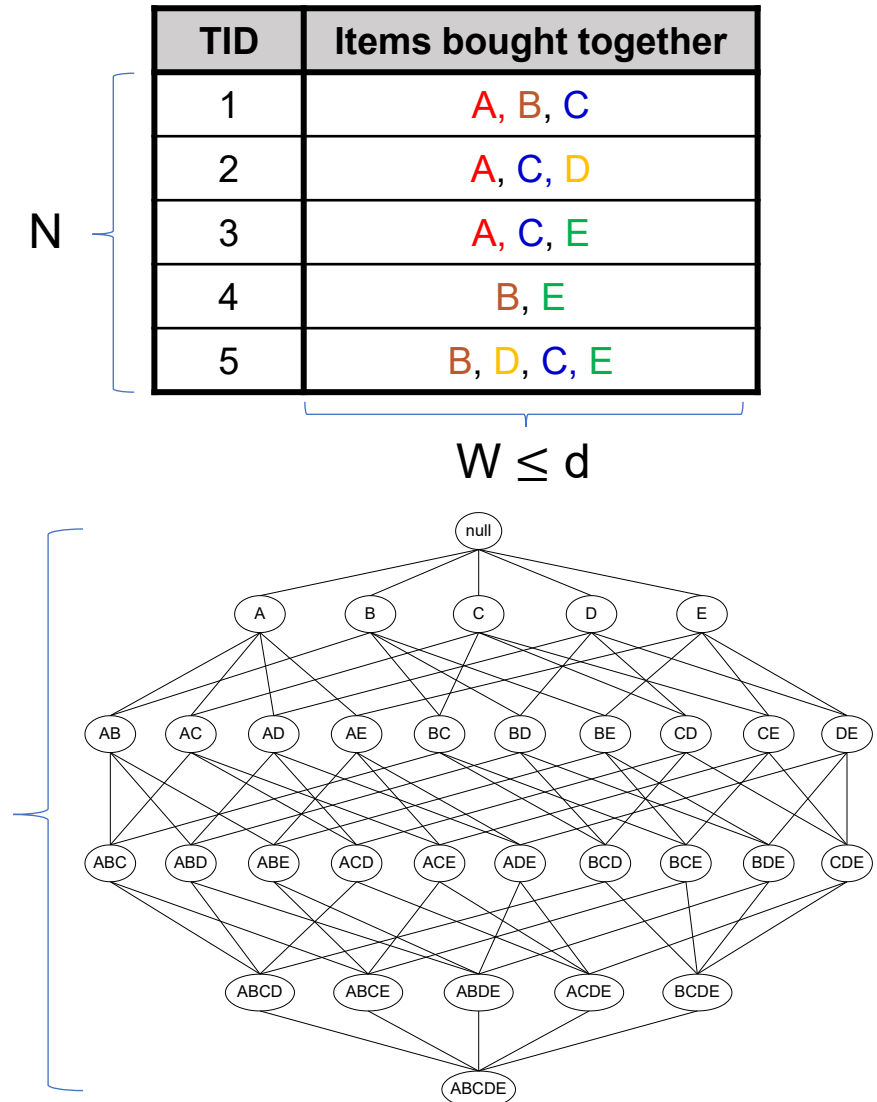
❑ Brute-force approach 2

■ Procedure

- Step 1: Find frequent itemsets (minsup)
 - From the unique items, generate all possible itemsets
 - Compute the support and prune by the minsup threshold
- Step 2: Find association rules (minconf)
 - From the frequent itemsets, generate all possible association rules
 - Compute the confidence and prune by the minconf threshold

■ Issue: step 1 is still computationally expensive!

- Given d unique items,
 - The number of transactions: N
 - The number of items in transactions: W
 - The number of itemsets: $M = 2^d$
- The computational complexity $\sim O(NWM)$
 - For each of N transactions with W items, check against M itemsets



Pattern Mining > Methodologies > Rule Discovery

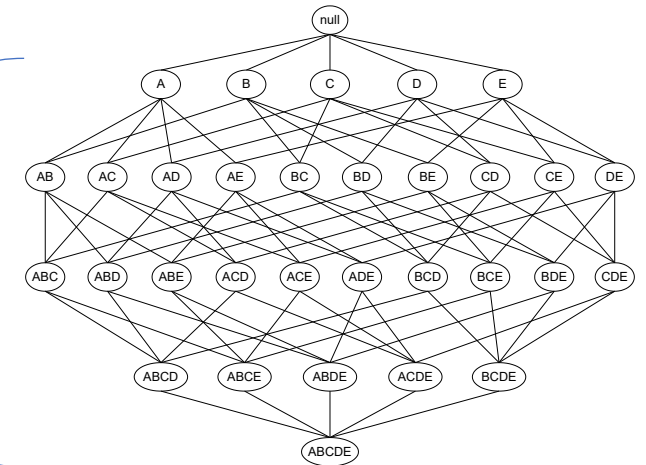
□ Strategies for finding frequent itemsets

- Reduce the number of transactions (N)
 - Reduce the size of N as the size of itemset increases
 - e.g., Partitioning, vertical-data format (i.e., item $\rightarrow$ TIDs)
- Reduce the number of candidates (M)
 - Use pruning techniques to reduce M
 - e.g., **Apriori**
- Reduce the number of comparisons ($N \times M$)
 - Use efficient data structures (e.g., hash tables) to store the candidates or transactions
 - No need to match every candidate against every transaction
 - e.g., Dynamic hashing and pruning, **FP-growth**

TID	Items bought together
1	A, B, C
2	A, C, D
3	A, C, E
4	B, E
5	B, D, C, E

$N \times M$

M



Outline

- ❑ Basic concepts of pattern mining
- ❑ **Pattern mining methodologies**
 - Association rule discovery
 - **Apriori**
 - FP-Growth
 - Rule generation
- ❑ Pattern mining evaluation

■ Pattern Mining > Methodologies > Apriori

□ Revisiting the extreme cases

TID	Items bought together
1	$a_1, a_2, \dots, a_{49}, a_{50}$
2	$a_1, a_2, \dots, a_{99}, a_{100}$

- Suppose that we have $\text{minsup} = 2$
- We have a closed and maximal frequent itemset: $\{a_1, \dots, a_{50}\}:2$
- *What are the supports of $\{a_1\}, \{a_2\}, \dots, \{a_{50}\}, \{a_1, a_2\}, \dots, \{a_1, \dots, a_{49}\}, \dots$?*
- *Can we guarantee that all subsets of $\{a_1, \dots, a_{50}\}$ are frequent?*
- *Can we guarantee that these are all frequent itemsets?*

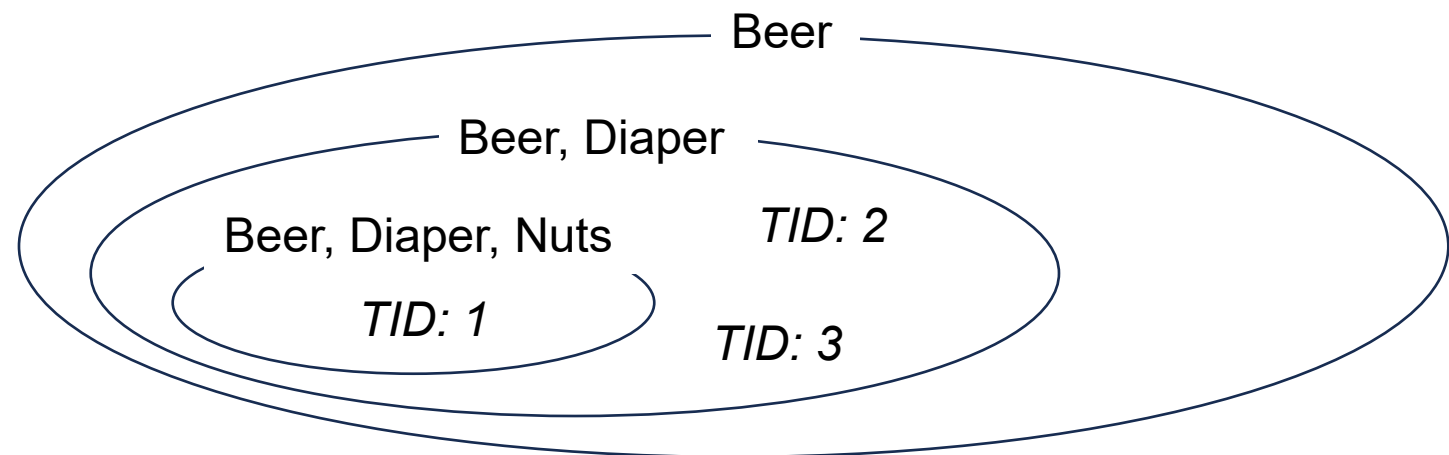
Pattern Mining > Methodologies > Apriori

□ Observation

- All items in an itemset X should be in its superset Y
 - e.g., every transaction containing $Y = \{\text{Beer, Diaper, Nuts}\}$ also contains $X = \{\text{Beer, Diaper}\}$
- The support of an itemset X is larger than or equal to the support of its superset Y
 - i.e., $\forall X, Y: (X \subseteq Y), s(X) \geq s(Y)$ a.k.a. **anti-monotone property of support**

TID	Items bought together
1	Beer, Nuts, Diaper
2	Beer, Coffee, Diaper
3	Beer, Diaper, Eggs
4	Nuts, Eggs, Milk
5	Nuts, Coffee, Diaper, Eggs, Milk

Sets of transactions containing item(s)



■ Pattern Mining > Methodologies > Apriori

□ Observation

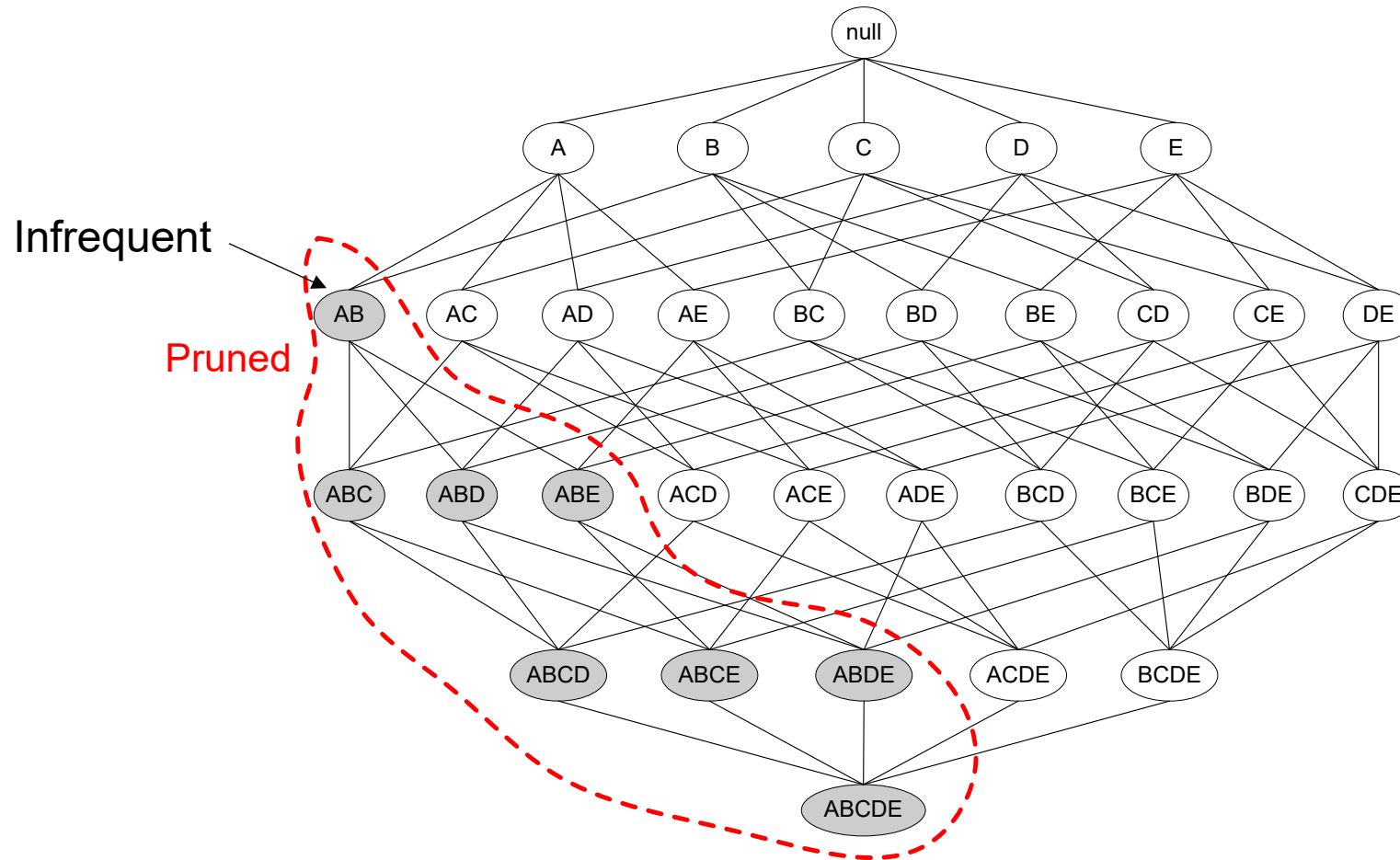
- All items in an itemset X should be in its superset Y
 - e.g., every transaction containing $Y = \{\text{Beer, Diaper, Nuts}\}$ also contains $X = \{\text{Beer, Diaper}\}$
- The support of an itemset X is larger than or equal to the support of its superset Y
 - i.e., $\forall X, Y: (X \subseteq Y), s(X) \geq s(Y)$ a.k.a. ***anti-monotone property of support***

□ The downward closure (a.k.a. “Apriori”) property of frequent patterns

- Apriori: a Latin phrase that means “*from what is earlier*”
- If an itemset Y is frequent, then all of its subsets X must also be frequent
 - i.e., $\forall X, Y: (X \subseteq Y), s(Y) \geq \text{minsup} \rightarrow s(X) \geq \text{minsup}$
- In other words, if an itemset X is not frequent, any superset Y of it can not be frequent
 - i.e., $\forall X, Y: (X \subseteq Y), s(X) < \text{minsup} \rightarrow s(Y) < \text{minsup}$

Pattern Mining > Methodologies > Apriori

- Example usage of the apriori property of frequent patterns
 - *If an itemset $\{A, B\}$ is infrequent, should we check all of its super-itemsets?*



■ Pattern Mining > Methodologies > Apriori

□ The Apriori algorithm

■ Basic idea

- From $k=1$, incrementally generate candidate itemsets and find frequent itemsets by scanning TDB
 - i.e., if k -itemset is frequent, $(k+1)$ -itemset can be frequent

■ Overall procedure for finding frequent itemsets

C_k : Candidate itemset of size k

F_k : Frequent itemset of size k

$k := 1$;

$F_k := \{\text{frequent items}\}$; // Frequent 1-itemset

While ($F_k \neq \emptyset$) **do** { // When F_k is non-empty

$C_{k+1} := \text{candidates generated from } F_k$; // candidate generation

 Derive F_{k+1} by counting candidates in C_{k+1} in TDB and compare with minsup;

$k := k + 1$

}

return $\cup_k F_k$ // Return F_k generated at each level

Pattern Mining > Methodologies > Apriori

□ Pseudo-code of Apriori for finding frequent itemsets

Input:

- D , a database of transactions;
- min_sup , the minimum support count threshold.

Output: L , frequent itemsets in D .

Method:

```
(1)  $L_1 = \text{find\_frequent\_1-itemsets}(D);$ 
(2) for ( $k = 2; L_{k-1} \neq \phi; k++$ ) {
(3)    $C_k = \text{apriori\_gen}(L_{k-1});$ 
(4)   for each transaction  $t \in D$  { // scan  $D$  for counts
(5)      $C_t = \text{subset}(C_k, t);$  // get the subsets of  $t$  that are candidates
(6)     for each candidate  $c \in C_t$ 
(7)        $c.\text{count}++;$ 
(8)   }
(9)    $L_k = \{c \in C_k | c.\text{count} \geq min\_sup\}$ 
(10) }
(11) return  $L = \cup_k L_k;$ 
```

procedure $\text{apriori_gen}(L_{k-1}:\text{frequent } (k-1)\text{-itemsets})$

```
(1)   for each itemset  $l_1 \in L_{k-1}$ 
(2)     for each itemset  $l_2 \in L_{k-1}$ 
(3)       if ( $(l_1[1] = l_2[1]) \wedge (l_1[2] = l_2[2])$ 
            $\wedge \dots \wedge (l_1[k-2] = l_2[k-2]) \wedge (l_1[k-1] < l_2[k-1])$ ) then {
(4)          $c = l_1 \bowtie l_2;$  // join step: generate candidates
(5)         if  $\text{has\_infrequent\_subset}(c, L_{k-1})$  then
(6)           delete  $c;$  // prune step: remove unfruitful candidate
(7)         else add  $c$  to  $C_k;$ 
(8)       }
(9)   return  $C_k;$ 
```

procedure $\text{has_infrequent_subset}(c:\text{candidate } k\text{-itemset};$

```
    $L_{k-1}:\text{frequent } (k-1)\text{-itemsets});$  // use prior knowledge
(1)   for each  $(k-1)$ -subset  $s$  of  $c$ 
(2)     if  $s \notin L_{k-1}$  then
(3)       return TRUE;
(4)   return FALSE;
```

■ Pattern Mining > Methodologies > Apriori

□ Case study

TDB

TID	Items bought together
1	Beer, Nuts, Diaper
2	Beer, Diaper, Coffee
3	Nuts, Diaper, Eggs, Milk
4	Nuts, Milk
5	Beer, Nuts, Diaper, Milk

minsup = 40% → 2

Pattern Mining > Methodologies > Apriori

□ Case study

Candidate
1-itemsets

C_1

Itemset
{Beer}
{Nuts}
{Diaper}
{Coffee}
{Eggs}
{Milk}

TDB

TID	Items bought together
1	Beer, Nuts, Diaper
2	Beer, Diaper, Coffee
3	Nuts, Diaper, Eggs, Milk
4	Nuts, Milk
5	Beer, Nuts, Diaper, Milk

minsup = 40% → 2

Pattern Mining > Methodologies > Apriori

□ Case study

Frequent 1-itemsets

C_1		F_1	
Itemset		Itemset	Sup
{Beer}		{Beer}	3
{Nuts}		{Nuts}	4
{Diaper}		{Diaper}	4
{Coffee}		{Coffee}	1
{Eggs}		{Eggs}	1
{Milk}		{Milk}	3

TDB

TID	Items bought together
1	Beer, Nuts, Diaper
2	Beer, Diaper, Coffee
3	Nuts, Diaper, Eggs, Milk
4	Nuts, Milk
5	Beer, Nuts, Diaper, Milk

minsup = 40% → 2

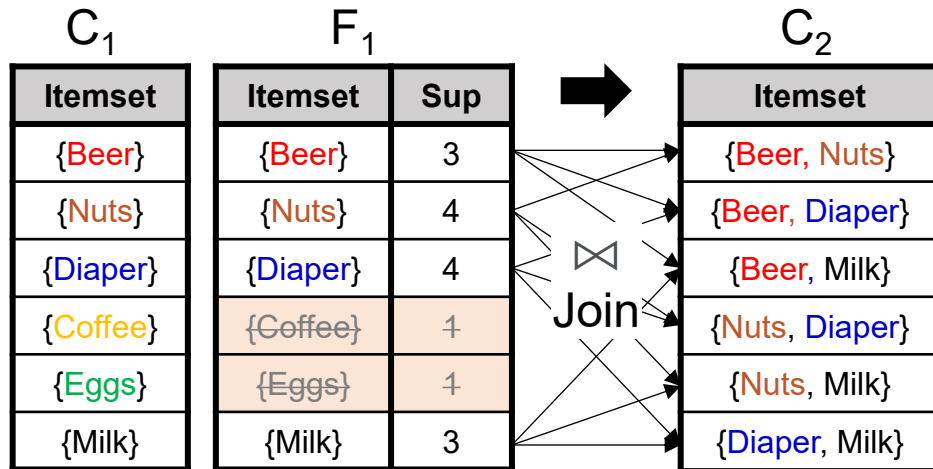
Pattern Mining > Methodologies > Apriori

□ Case study

TDB

TID	Items bought together
1	Beer, Nuts, Diaper
2	Beer, Diaper, Coffee
3	Nuts, Diaper, Eggs, Milk
4	Nuts, Milk
5	Beer, Nuts, Diaper, Milk

minsup = 40% → 2



Pattern Mining > Methodologies > Apriori

□ Case study

TDB

TID	Items bought together
1	Beer, Nuts, Diaper
2	Beer, Diaper, Coffee
3	Nuts, Diaper, Eggs, Milk
4	Nuts, Milk
5	Beer, Nuts, Diaper, Milk

minsup = 40% → 2

C ₁			F ₁						C ₂			F ₂		
Itemset	Itemset	Sup	Itemset	Sup		Itemset	Sup		Itemset	Sup		Itemset	Sup	
{Beer}	{Beer}	3	{Beer}	3	→	{Beer, Nuts}	2		{Beer, Nuts}	2		{Beer, Nuts}	2	
{Nuts}	{Nuts}	4	{Nuts}	4		{Beer, Diaper}	3		{Beer, Diaper}	3		{Beer, Diaper}	3	
{Diaper}	{Diaper}	4	{Diaper}	4		{Beer, Milk}	4		{Beer, Milk}	4		{Beer, Milk}	4	
{Coffee}	{Coffee}	1	{Coffee}	1		{Nuts, Diaper}	3		{Nuts, Diaper}	3		{Nuts, Diaper}	3	
{Eggs}	{Eggs}	1	{Eggs}	1		{Nuts, Milk}	3		{Nuts, Milk}	3		{Nuts, Milk}	3	
{Milk}	{Milk}	3	{Milk}	3		{Diaper, Milk}	2		{Diaper, Milk}	2		{Diaper, Milk}	2	

Pattern Mining > Methodologies > Apriori

□ Case study

TDB

TID	Items bought together
1	Beer, Nuts, Diaper
2	Beer, Diaper, Coffee
3	Nuts, Diaper, Eggs, Milk
4	Nuts, Milk
5	Beer, Nuts, Diaper, Milk

minsup = 40% → 2

C ₁			F ₁						C ₂			F ₂						C ₃		
Itemset	Itemset	Sup	Itemset	Sup		Itemset	Sup		Itemset	Sup		Itemset	Sup		Itemset	Sup		Itemset	Sup	
{Beer}	{Beer}	3	{Beer}	3	→	{Beer, Nuts}	2	Join	{Beer, Nuts}	2	→	{Beer, Nuts, Diaper}		Join	{Beer, Nuts, Diaper}		Join	{Beer, Nuts, Diaper}		Join
{Nuts}	{Nuts}	4	{Nuts}	4		{Beer, Diaper}	3		{Beer, Diaper}	3		{Beer, Nuts, Milk}			{Beer, Nuts, Milk}			{Beer, Nuts, Milk}		
{Diaper}	{Diaper}	4	{Diaper}	4		{Beer, Milk}	4		{Beer, Milk}	4		{Beer, Diaper, Milk}			{Beer, Diaper, Milk}			{Beer, Diaper, Milk}		
{Coffee}	{Coffee}	1	{Coffee}	1		{Nuts, Diaper}	3		{Nuts, Diaper}	3		{Nuts, Diaper, Milk}			{Nuts, Diaper, Milk}			{Nuts, Diaper, Milk}		
{Eggs}	{Eggs}	1	{Eggs}	1		{Nuts, Milk}	3		{Nuts, Milk}	3										
{Milk}	{Milk}	3	{Milk}	3		{Diaper, Milk}	2		{Diaper, Milk}	2										

Pattern Mining > Methodologies > Apriori

□ Case study

TDB

TID	Items bought together
1	Beer, Nuts, Diaper
2	Beer, Diaper, Coffee
3	Nuts, Diaper, Eggs, Milk
4	Nuts, Milk
5	Beer, Nuts, Diaper, Milk

minsup = 40% → 2

C_1 F_1

Itemset	Itemset	Sup
{Beer}	{Beer}	3
{Nuts}	{Nuts}	4
{Diaper}	{Diaper}	4
{Coffee}	{Coffee}	4
{Eggs}	{Eggs}	4
{Milk}	{Milk}	3



C_2 F_2

Itemset	Itemset	Sup
{Beer, Nuts}	{Beer, Nuts}	2
{Beer, Diaper}	{Beer, Diaper}	3
{Beer, Milk}	{Beer, Milk}	4
{Nuts, Diaper}	{Nuts, Diaper}	3
{Nuts, Milk}	{Nuts, Milk}	3
{Diaper, Milk}	{Diaper, Milk}	2



C_3

Itemset
{Beer, Nuts, Diaper}
{Beer, Nuts, Milk}
{Beer, Diaper, Milk}
{Nuts, Diaper, Milk}

*Should we count
all of them?*

Pattern Mining > Methodologies > Apriori

□ Case study

TDB

TID	Items bought together
1	Beer, Nuts, Diaper
2	Beer, Diaper, Coffee
3	Nuts, Diaper, Eggs, Milk
4	Nuts, Milk
5	Beer, Nuts, Diaper, Milk

minsup = 40% → 2

C_1 F_1

Itemset	Itemset	Sup
{Beer}	{Beer}	3
{Nuts}	{Nuts}	4
{Diaper}	{Diaper}	4
{Coffee}	{Coffee}	4
{Eggs}	{Eggs}	4
{Milk}	{Milk}	3



C_2 F_2

Itemset	Itemset	Sup
{Beer, Nuts}	{Beer, Nuts}	2
{Beer, Diaper}	{Beer, Diaper}	3
{Beer, Milk}	{Beer, Milk}	4
{Nuts, Diaper}	{Nuts, Diaper}	3
{Nuts, Milk}	{Nuts, Milk}	3
{Diaper, Milk}	{Diaper, Milk}	2



C_3

Itemset
{Beer, Nuts, Diaper}
{Beer, Nuts, Milk}
{Beer, Diaper, Milk}
{Nuts, Diaper, Milk}

No!

Should we count all of them?

Pattern Mining > Methodologies > Apriori

□ Case study

TDB

TID	Items bought together
1	Beer, Nuts, Diaper
2	Beer, Diaper, Coffee
3	Nuts, Diaper, Eggs, Milk
4	Nuts, Milk
5	Beer, Nuts, Diaper, Milk

minsup = 40% → 2

C ₁			F ₁			C ₂			F ₂			C ₃			F ₃		
Itemset	Itemset	Sup	Itemset	Itemset	Sup	Itemset	Itemset	Sup	Itemset	Itemset	Sup	Itemset	Itemset	Sup	Itemset	Itemset	Sup
{Beer}	{Beer}	3	{Beer}	{Beer}	3	{Beer, Nuts}	{Beer, Nuts}	2	{Beer, Nuts}	{Beer, Nuts}	2	{Beer, Nuts, Diaper}	{Beer, Nuts, Diaper}	2	{Beer, Nuts, Diaper}	{Beer, Nuts, Diaper}	2
{Nuts}	{Nuts}	4	{Nuts}	{Nuts}	4	{Beer, Diaper}	{Beer, Diaper}	3	{Beer, Diaper}	{Beer, Diaper}	3	{Beer, Nuts, Milk}	{Beer, Nuts, Milk}		{Beer, Nuts, Milk}	{Beer, Nuts, Milk}	
{Diaper}	{Diaper}	4	{Diaper}	{Diaper}	4	{Beer, Milk}	{Beer, Milk}	4	{Beer, Milk}	{Beer, Milk}		{Beer, Diaper, Milk}	{Beer, Diaper, Milk}		{Beer, Diaper, Milk}	{Beer, Diaper, Milk}	
{Coffee}	{Coffee}	1	{Coffee}	{Coffee}		{Nuts, Diaper}	{Nuts, Diaper}	3	{Nuts, Diaper}	{Nuts, Diaper}	3	{Nuts, Diaper, Milk}	{Nuts, Diaper, Milk}	2	{Nuts, Diaper, Milk}	{Nuts, Diaper, Milk}	2
{Eggs}	{Eggs}	1	{Eggs}	{Eggs}		{Nuts, Milk}	{Nuts, Milk}	3	{Nuts, Milk}	{Nuts, Milk}	3						
{Milk}	{Milk}	3	{Milk}	{Milk}	3	{Diaper, Milk}	{Diaper, Milk}	2	{Diaper, Milk}	{Diaper, Milk}	2						

Pattern Mining > Methodologies > Apriori

□ Case study

TDB

TID	Items bought together
1	Beer, Nuts, Diaper
2	Beer, Diaper, Coffee
3	Nuts, Diaper, Eggs, Milk
4	Nuts, Milk
5	Beer, Nuts, Diaper, Milk

minsup = 40% → 2

C ₁		
Itemset	Itemset	Sup
{Beer}	{Beer}	3
{Nuts}	{Nuts}	4
{Diaper}	{Diaper}	4
{Coffee}	{Coffee}	1
{Eggs}	{Eggs}	1
{Milk}	{Milk}	3



C ₂		
Itemset	Itemset	Sup
{Beer, Nuts}	{Beer, Nuts}	2
{Beer, Diaper}	{Beer, Diaper}	3
{Beer, Milk}	{Beer, Milk}	1
{Nuts, Diaper}	{Nuts, Diaper}	3
{Nuts, Milk}	{Nuts, Milk}	3
{Diaper, Milk}	{Diaper, Milk}	2



C ₃		
Itemset	Itemset	Sup
{Beer, Nuts, Diaper}	{Beer, Nuts, Diaper}	2
{Beer, Nuts, Milk}	{Beer, Nuts, Milk}	1
{Beer, Diaper, Milk}	{Beer, Diaper, Milk}	1
{Nuts, Diaper, Milk}	{Nuts, Diaper, Milk}	2

F ₃		
Itemset	Itemset	Sup
{Beer, Nuts, Diaper}	{Beer, Nuts, Diaper}	2
{Nuts, Diaper, Milk}	{Nuts, Diaper, Milk}	2

⋈ Join

C₄

Itemset	Sup
{Beer, Nuts, Diaper, Milk}	

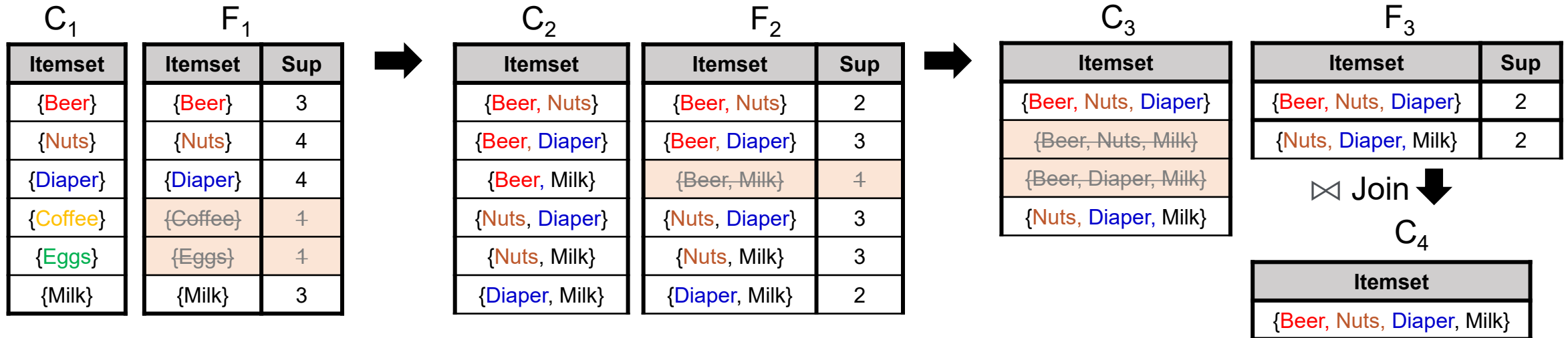
Pattern Mining > Methodologies > Apriori

□ Case study

TDB

TID	Items bought together
1	Beer, Nuts, Diaper
2	Beer, Diaper, Coffee
3	Nuts, Diaper, Eggs, Milk
4	Nuts, Milk
5	Beer, Nuts, Diaper, Milk

minsup = 40% → 2



Is {Beer, Nuts, Milk} frequent?
Is {Beer, Diaper, Milk} frequent?

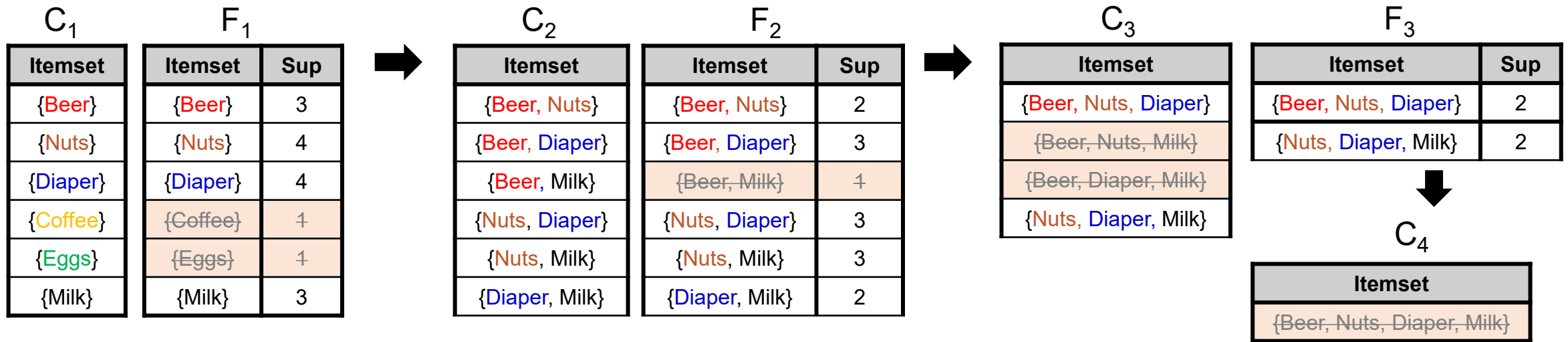
Pattern Mining > Methodologies > Apriori

□ Case study

TDB

TID	Items bought together
1	Beer, Nuts, Diaper
2	Beer, Diaper, Coffee
3	Nuts, Diaper, Eggs, Milk
4	Nuts, Milk
5	Beer, Nuts, Diaper, Milk

minsup = 40% → 2



If we consider every possible itemsets → $2^6 - 1 = 63$ scans

With Apriori → $6 + 6 + 2 = 14$ scans

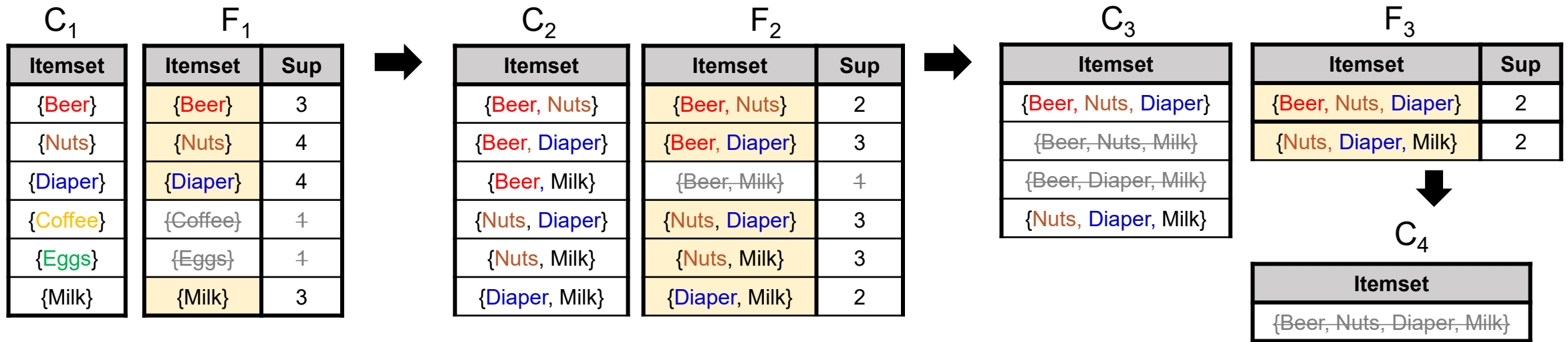
Pattern Mining > Methodologies > Apriori

□ Case study

TDB

TID	Items bought together
1	Beer, Nuts, Diaper
2	Beer, Diaper, Coffee
3	Nuts, Diaper, Eggs, Milk
4	Nuts, Milk
5	Beer, Nuts, Diaper, Milk

minsup = 40% → 2



We have found 11 frequent itemsets

Pattern Mining > Methodologies > Apriori

□ Quiz: how many frequent k-itemsets ($k > 1$) are there?

TDB

TID	Items bought together
1	Beer, Coffee, Nuts, Diaper, Milk
2	Beer, Coffee, Diaper
3	Beer, Nuts, Diaper, Eggs
4	Nuts, Diaper, Milk
5	Beer, Nuts, Diaper, Milk

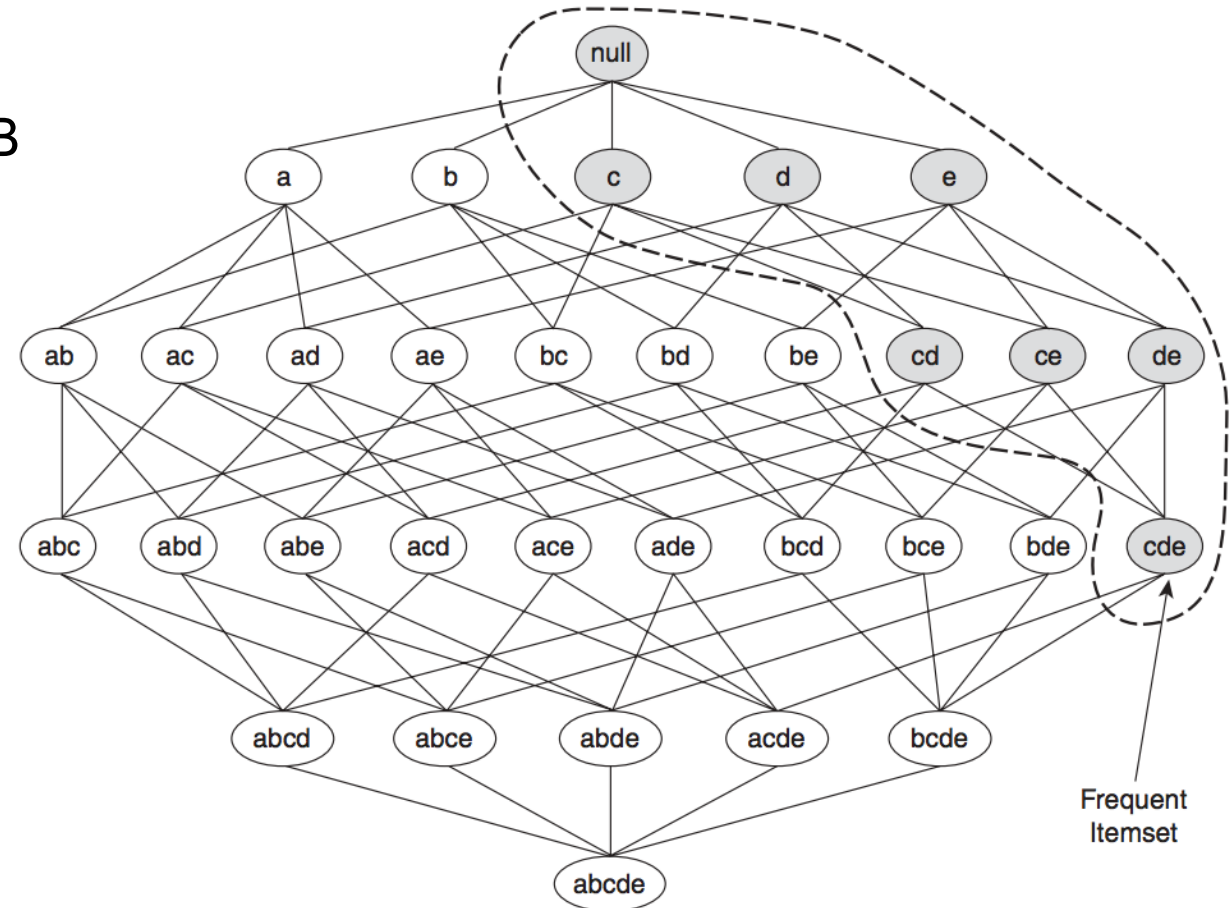
minsup = 60% → 3

C ₁			F ₁			C ₂			F ₂			C ₃			F ₃			C ₄			F ₄		
Itemset	Itemset	Sup	Itemset	Sup		Itemset	Itemset	Sup	Itemset	Itemset	Sup	Itemset	Itemset	Sup	Itemset	Itemset	Sup	Itemset	Itemset	Sup	Itemset	Itemset	Sup
{Beer}	{B}	4																					
{Nuts}	{N}	4																					
{Diaper}	{D}	5																					
{Coffee}	{C}	2																					
{Eggs}	{E}	1																					
{Milk}	{M}	3																					

Pattern Mining > Methodologies > Apriori

□ Apriori: a breadth-first search algorithm

- For each k , derive candidate $(k+1)$ -itemsets
- Find the frequent $(k+1)$ -itemsets from the TDB



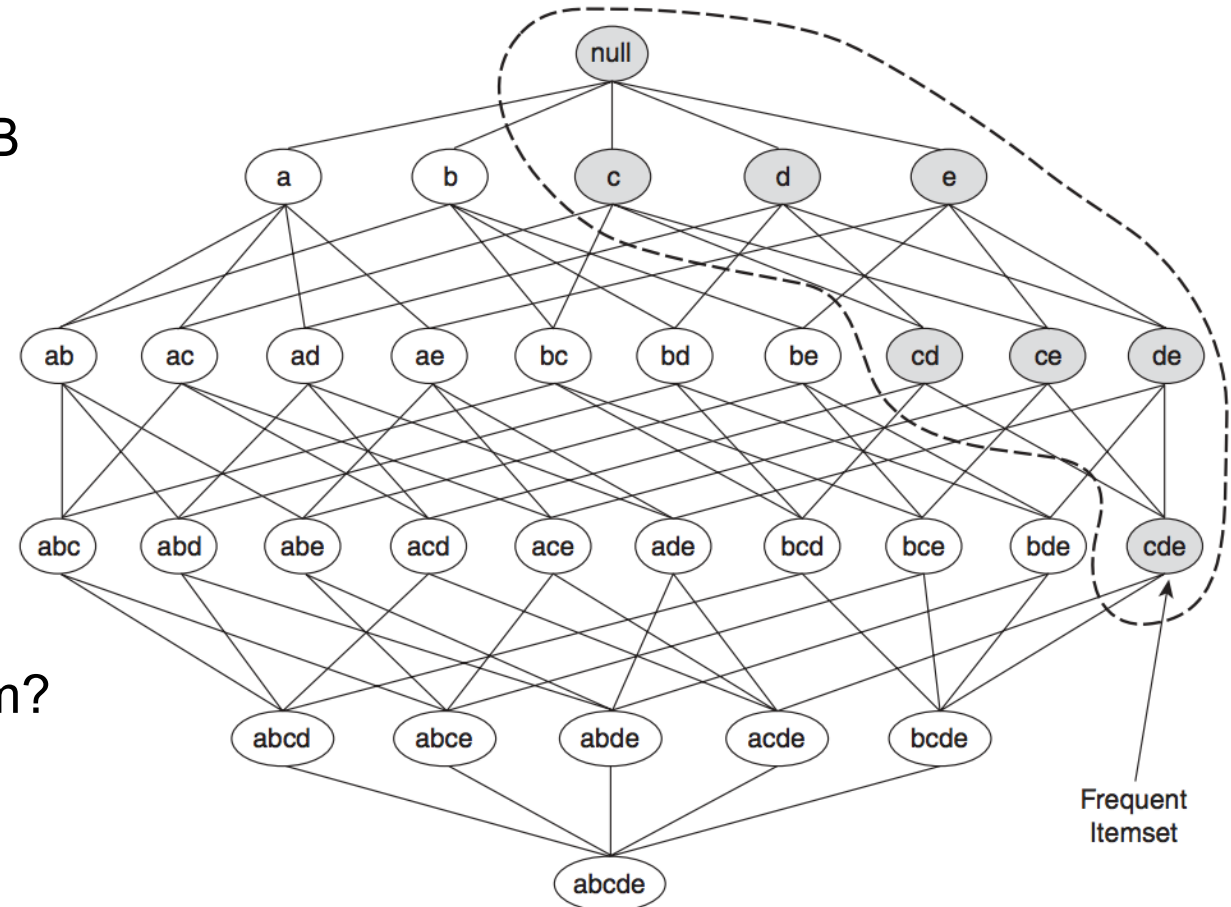
Pattern Mining > Methodologies > Apriori

□ Apriori: a breadth-first search algorithm

- For each k , derive candidate $(k+1)$ -itemsets
- Find the frequent $(k+1)$ -itemsets from the TDB

□ Questions

- Should we always scan the entire TDB to check if an itemset is frequent?
- Can we develop a depth-first search algorithm?



Outline

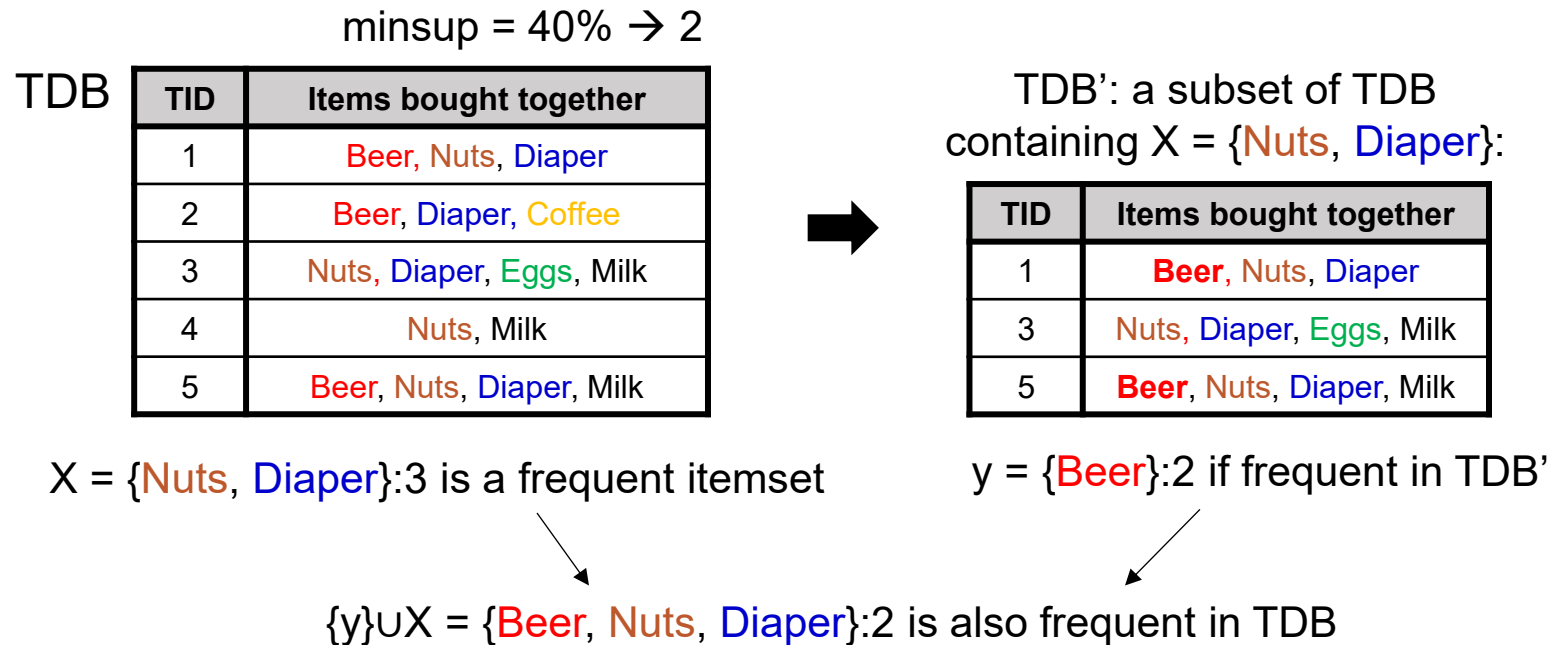
- ❑ Basic concepts of pattern mining
- ❑ **Pattern mining methodologies**
 - Association rule discovery
 - Apriori
 - **FP-Growth**
 - Rule generation
- ❑ Pattern mining evaluation

Pattern Mining > Methodologies > FP-Growth

□ Motivation

- In TDB, let X be a frequent itemset, TDB' be the set of transactions that contain X
- If a new item y is a frequent item in TDB' , then $\{y\} \cup X$ must be a frequent itemset

▪ Example



■ Pattern Mining > Methodologies > FP-Growth

□ Motivation

- In TDB, let X be a frequent itemset, TDB' be the set of transactions that contain X
- If a new item y is a frequent item in TDB' , then $\{y\} \cup X$ must be a frequent itemset

□ FP-Growth (frequent pattern growth)

- Main idea
 - A compact data structure FP-tree is constructed to store information for finding frequent patterns
 - Recursively find frequent itemsets from FP-tree
- Advantages
 - No candidate itemsets generation is required (unlike Apriori)
 - **Scan TDB only twice!**

■ Pattern Mining > Methodologies > FP-Growth

□ The FP-Growth algorithm for finding frequent patterns

- Input: a transactional database TDB, minsup
- Procedures
 1. Find frequent 1-itemsets (1st TDB scan)
 - e.g., minsup = 60% (=3) → f:4, c:4, a:3, b:3, m:3, p:3

minsup = 60% → 3

TDB	TID	Items bought together
	1	<i>f, a, c, d, g, i, m, p</i>
	2	<i>a, b, c, f, l, m, o</i>
	3	<i>b, f, h, j, o, w</i>
	4	<i>b, c, k, s, p</i>
	5	<i>a, f, c, e, l, p, m, n</i>

Support of 1-itemsets

f:4, c:4, a:3, b:3, m:3, p:3, l:2, o:2, d:1, e:1, g:1, h:1, i:1, k:1, n:1, s:1, w:1

Frequent 1-itemsets

f:4, c:4, a:3, b:3, m:3, p:3

■ Pattern Mining > Methodologies > FP-Growth

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- Input: a transactional database TDB, minsup
- Procedures
 1. Find frequent 1-itemsets (1st TDB scan)
 - e.g., minsup = 60% (=3) → f:4, c:4, a:3, b:3, m:3, p:3
 2. Build f-lists
 - For each transactions, sort the frequent items by frequency to get a f-list
 - e.g., {f, a, c, d, g, i, m, p} → f, c, a, m, p

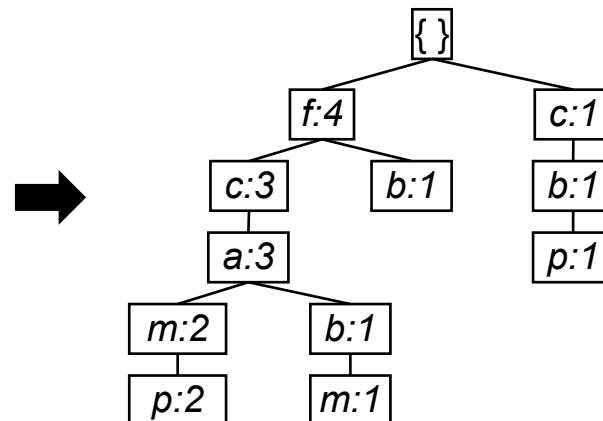
TID	Items bought together	f-list
1	<i>f, a, c, d, g, i, m, p</i>	<i>f, c, a, m, p</i>
2	<i>a, b, c, f, l, m, o</i>	<i>f, c, a, b, m</i>
3	<i>b, f, h, j, o, w</i>	<i>f, b</i>
4	<i>b, c, k, s, p</i>	<i>c, b, p</i>
5	<i>a, f, c, e, l, p, m, n</i>	<i>f, c, a, m, p</i>

Pattern Mining > Methodologies > FP-Growth

□ The FP-Growth algorithm for finding frequent patterns

- Input: a transactional database TDB, minsup
- Procedures
 1. Find frequent 1-itemsets (1st TDB scan)
 2. Build f-lists
 - 3. Construct FP-tree (2nd TDB scan)**
 - 4. Find frequent itemsets with FP-tree**

TID	Items bought together	f-list
1	<i>f, a, c, d, g, i, m, p</i>	<i>f, c, a, m, p</i>
2	<i>a, b, c, f, l, m, o</i>	<i>f, c, a, b, m</i>
3	<i>b, f, h, j, o, w</i>	<i>f, b</i>
4	<i>b, c, k, s, p</i>	<i>c, b, p</i>
5	<i>a, f, c, e, l, p, m, n</i>	<i>f, c, a, m, p</i>



	Frequent itemset
f:4	None
c:4	{f, c}:3
a:3	{f, a}:3, {f, c}:3, {f, c, a}:3
b:3	None
m:3	{f, m}:3, {c, m}:3, {a, m}: 3, {f, c, m}:3, {f, a, m}:3, {c, a, m}:3, {f, c, a, m}:3
p:3	{c, p}: 3

Pattern Mining > Methodologies > FP-Growth

□ FP-Growth Step 3: construct **FP-Tree**

- Combine f-lists, with shared sub-branches merged, counts accumulated

FP-Tree

{ }

TDB with f-list (minsup=3)

TID	Items bought together	f-list
1	<i>f, a, c, d, g, i, m, p</i>	<i>f, c, a, m, p</i>
2	<i>a, b, c, f, l, m, o</i>	<i>f, c, a, b, m</i>
3	<i>b, f, h, j, o, w</i>	<i>f, b</i>
4	<i>b, c, k, s, p</i>	<i>c, b, p</i>
5	<i>a, f, c, e, l, p, m, n</i>	<i>f, c, a, m, p</i>

Header Table

Item	Support	Head
f	4	
c	4	
a	3	
b	3	
m	3	
p	3	

Pattern Mining > Methodologies > FP-Growth

□ FP-Growth Step 3: construct **FP-Tree**

- Combine f-lists, with shared sub-branches merged, counts accumulated

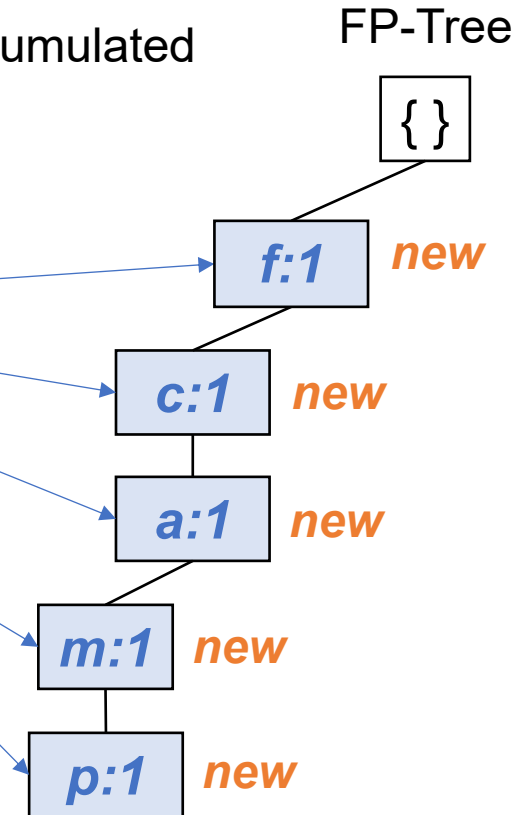
TDB with f-list (minsup=3)

TID	Items bought together	f-list
1	<i>f, a, c, d, g, i, m, p</i>	<i>f, c, a, m, p</i>
2	<i>a, b, c, f, l, m, o</i>	<i>f, c, a, b, m</i>
3	<i>b, f, h, j, o, w</i>	<i>f, b</i>
4	<i>b, c, k, s, p</i>	<i>c, b, p</i>
5	<i>a, f, c, e, l, p, m, n</i>	<i>f, c, a, m, p</i>



Header Table

Item	Support	Head
f	4	
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a	3	
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m	3	
p	3	



Pattern Mining > Methodologies > FP-Growth

□ FP-Growth Step 3: construct **FP-Tree**

- Combine f-lists, with shared sub-branches merged, counts accumulated

TDB with f-list (minsup=3)

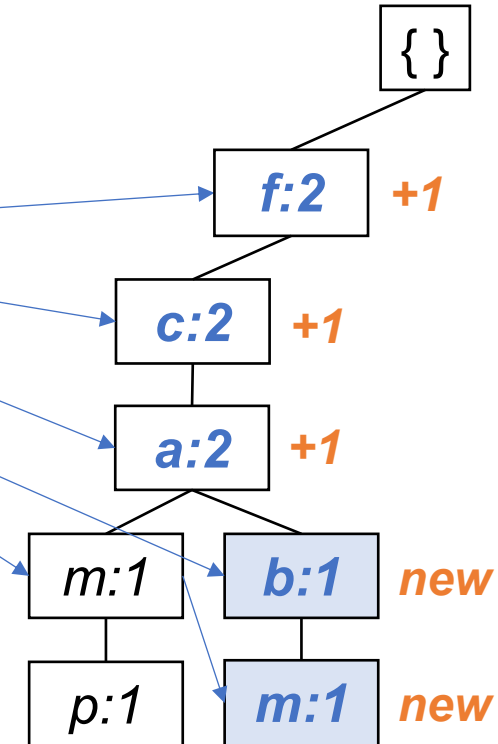
TID	Items bought together	f-list
1	<i>f, a, c, d, g, i, m, p</i>	<i>f, c, a, m, p</i>
2	<i>a, b, c, f, l, m, o</i>	<i>f, c, a, b, m</i>
3	<i>b, f, h, j, o, w</i>	<i>f, b</i>
4	<i>b, c, k, s, p</i>	<i>c, b, p</i>
5	<i>a, f, c, e, l, p, m, n</i>	<i>f, c, a, m, p</i>



Header Table

Item	Support	Head
f	4	
c	4	
a	3	
b	3	
m	3	
p	3	

FP-Tree



Pattern Mining > Methodologies > FP-Growth

□ FP-Growth Step 3: construct **FP-Tree**

- Combine f-lists, with shared sub-branches merged, counts accumulated

TDB with f-list (minsup=3)

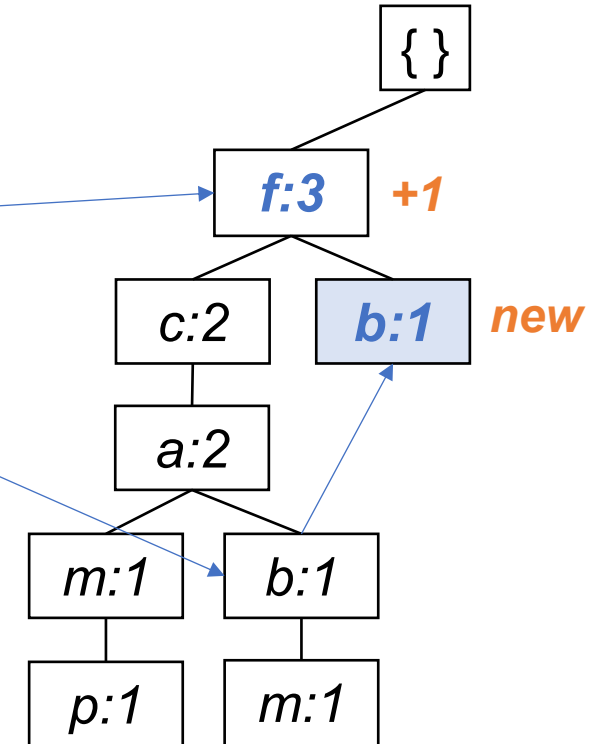
TID	Items bought together	f-list
1	<i>f, a, c, d, g, i, m, p</i>	<i>f, c, a, m, p</i>
2	<i>a, b, c, f, l, m, o</i>	<i>f, c, a, b, m</i>
3	<i>b, f, h, j, o, w</i>	<i>f, b</i>
4	<i>b, c, k, s, p</i>	<i>c, b, p</i>
5	<i>a, f, c, e, l, p, m, n</i>	<i>f, c, a, m, p</i>



Header Table

Item	Support	Head
f	4	
c	4	
a	3	
b	3	
m	3	
p	3	

FP-Tree



Pattern Mining > Methodologies > FP-Growth

□ FP-Growth Step 3: construct **FP-Tree**

- Combine f-lists, with shared sub-branches merged, counts accumulated

TDB with f-list (minsup=3)

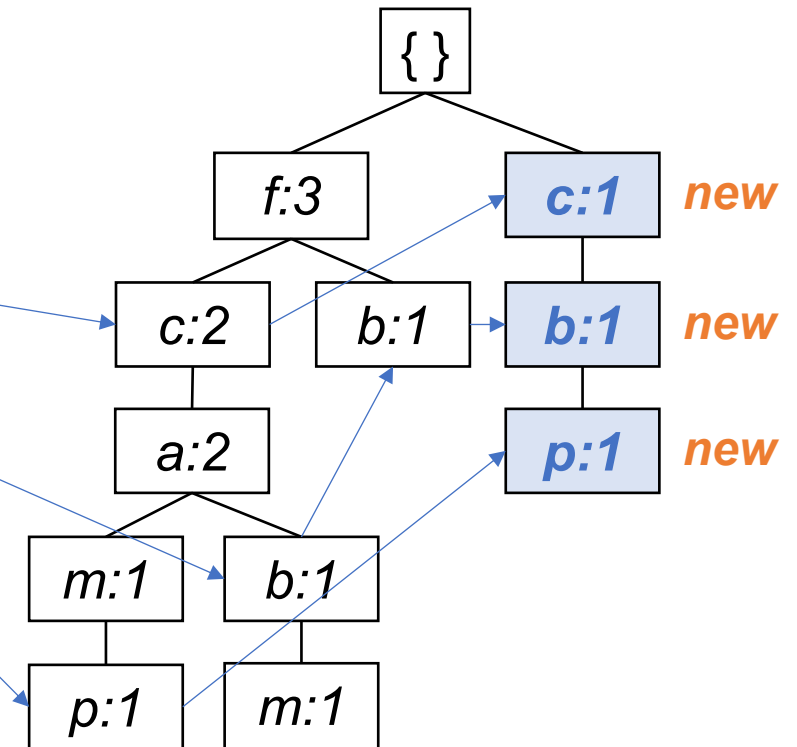
TID	Items bought together	f-list
1	<i>f, a, c, d, g, i, m, p</i>	<i>f, c, a, m, p</i>
2	<i>a, b, c, f, l, m, o</i>	<i>f, c, a, b, m</i>
3	<i>b, f, h, j, o, w</i>	<i>f, b</i>
4	<i>b, c, k, s, p</i>	<i>c, b, p</i>
5	<i>a, f, c, e, l, p, m, n</i>	<i>f, c, a, m, p</i>



Header Table

Item	Support	Head
f	4	
c	4	
a	3	
b	3	
m	3	
p	3	

FP-Tree



Pattern Mining > Methodologies > FP-Growth

□ FP-Growth Step 3: construct **FP-Tree**

- Combine f-lists, with shared sub-branches merged, counts accumulated

TDB with f-list (minsup=3)

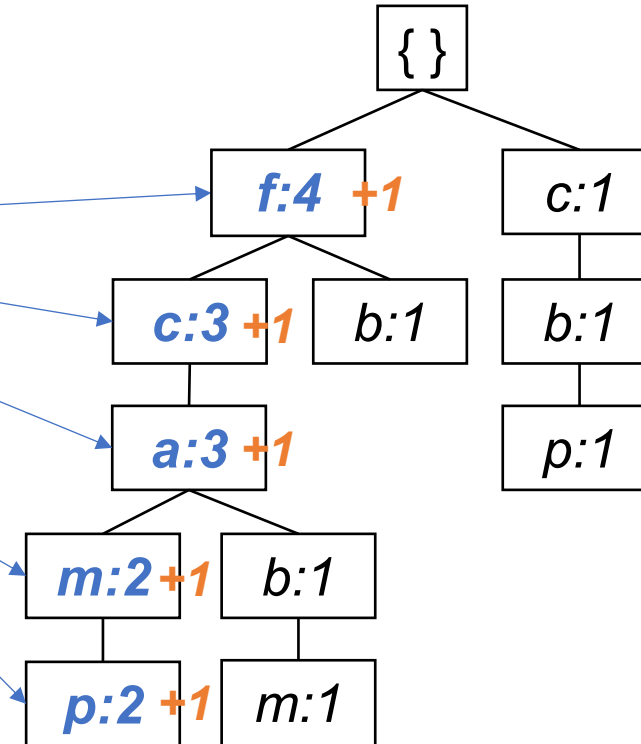
TID	Items bought together	f-list
1	<i>f, a, c, d, g, i, m, p</i>	<i>f, c, a, m, p</i>
2	<i>a, b, c, f, l, m, o</i>	<i>f, c, a, b, m</i>
3	<i>b, f, h, j, o, w</i>	<i>f, b</i>
4	<i>b, c, k, s, p</i>	<i>c, b, p</i>
5	<i>a, f, c, e, l, p, m, n</i>	<i>f, c, a, m, p</i>



Header Table

Item	Support	Head
f	4	
c	4	
a	3	
b	3	
m	3	
p	3	

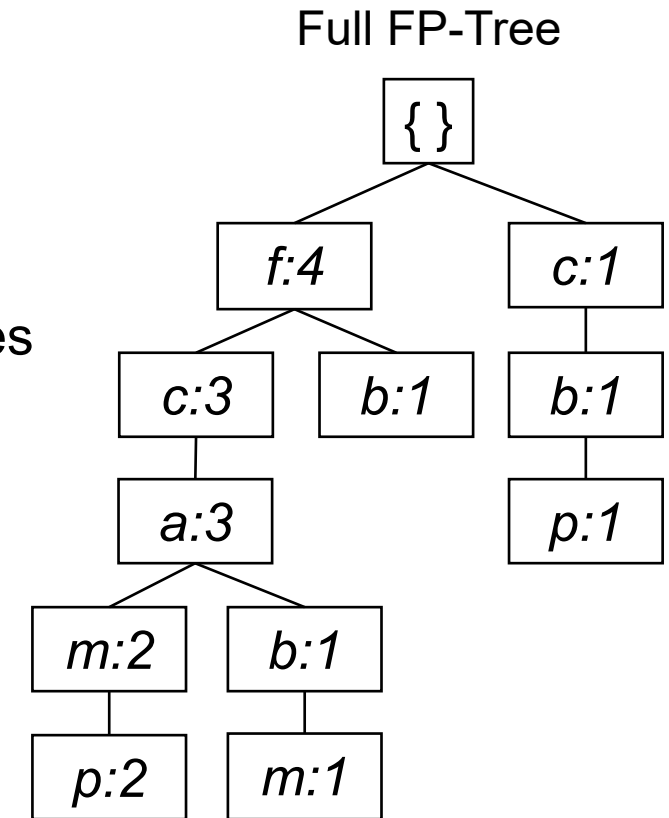
FP-Tree



Pattern Mining > Methodologies > FP-Growth

□ FP-Growth Step 4: find frequent itemsets with FP-Tree

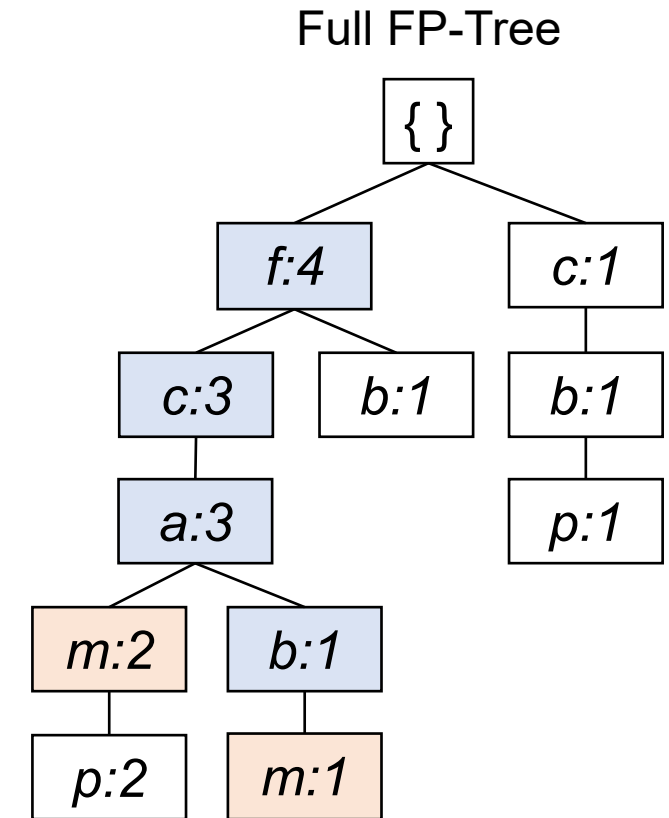
- 4.1 Form conditional pattern bases: **pattern_base** | α
- 4.2 Construct conditional FP-trees: **FP-tree** | α
- 4.3 Recursively find frequent itemsets from the conditional FP-trees



Pattern Mining > Methodologies > FP-Growth

□ FP-Growth Step 4.1: form conditional pattern bases: **pattern_base** | α

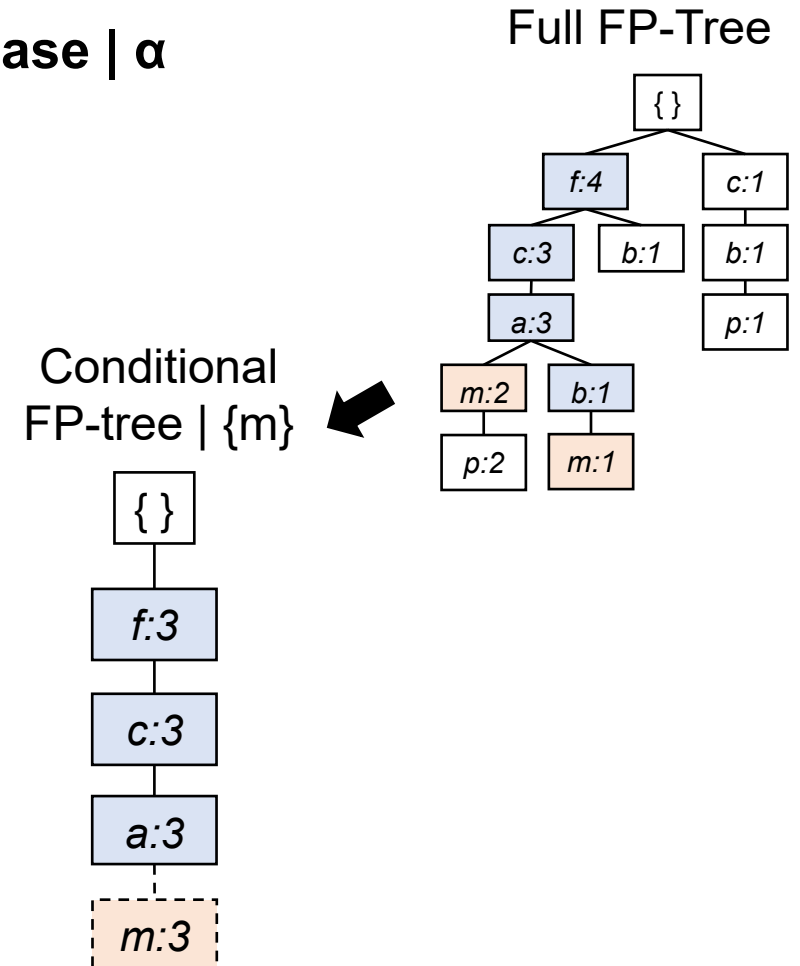
- A partition of frequent patterns
 - A sub TDB which consists of the set of prefix paths in the FP-tree with a frequent itemset α as a co-occurring suffix pattern
- e.g., $\alpha = \{m\}$
 - conditional pattern bases | $\{m\} = \{f,c,a\}:2, \{f,c,a,b\}:1$
 - i.e., $\{f, c, a\}$: support = 2 and $\{f, c, a, b\}$: support = 1



Pattern Mining > Methodologies > FP-Growth

□ FP-Growth Step 4.2: construct conditional FP-trees: **FP-tree** | α

- A small FP-tree on the conditional pattern base **pattern_base** | α
- e.g., $\alpha = \{m\}$
 - Conditional pattern bases | $\{m\} = \{f, c, a\}:2, \{f, c, a, b\}:1$
 - Conditional FP-tree | $\{m\} = f:3 - c:3 - a:3$
 - A conditional FP-tree with the given minsup = 3
 - $b:1 < \text{minsup} = 3$ is not included in the conditional FP-tree



Pattern Mining > Methodologies > FP-Growth

□ FP-Growth Step 4.3: recursively find frequent k-itemsets ($k > 1$)

- Recursively construct conditional FP-tree | α and find frequent itemsets

- e.g., $\alpha = \{m\}$

→ Conditional pattern bases | $\{m\} = \{f, c, a\}:2, \{f, c, a, b\}:1$

→ Conditional FP-tree | $\{m\} = f:3 - c:3 - a:3$

→ **Frequent 2-itemset:** $\{a, m\}, \{c, m\}, \{f, m\}$

→ Conditional pattern bases | $\{a, m\} = \{f, c\}:3$

→ Conditional FP-tree | $\{a, m\} = f:3 - c:3$

→ **Frequent 3-itemset:** $\{c, a, m\}, \{f, a, m\}$

→ Conditional pattern bases | $\{c, a, m\} = \{f\}:3$

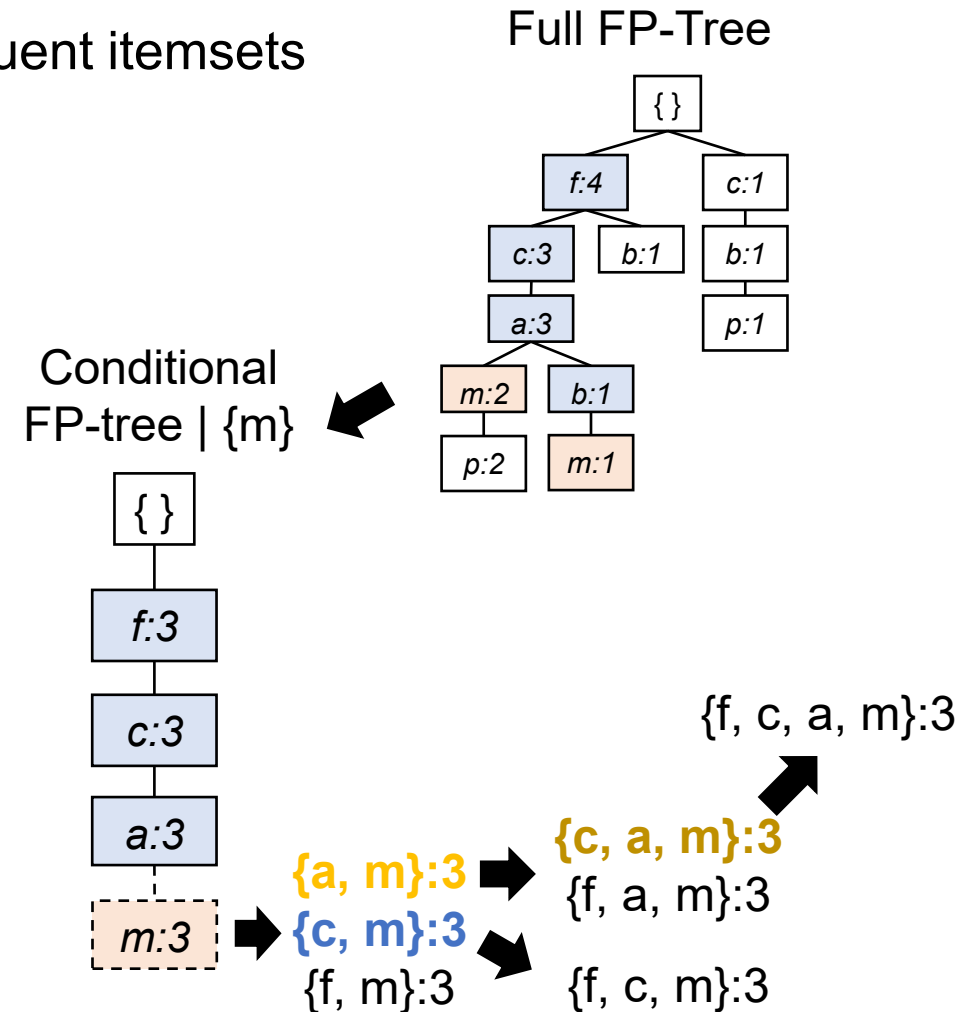
→ Conditional FP-tree | $\{c, a, m\} = f:3$

→ **Frequent 4-itemset:** $\{f, c, a, m\}$

→ Conditional pattern bases | $\{c, m\} = \{f\}:3$

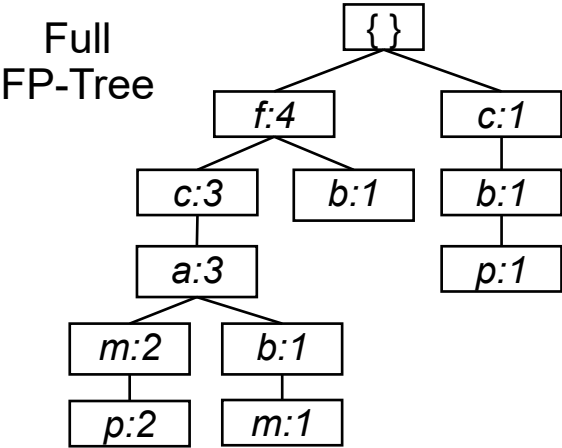
→ Conditional FP-tree | $\{c, m\} = f:3$

→ **Frequent 3-itemset:** $\{f, c, m\}$



Pattern Mining > Methodologies > FP-Growth

- Example of FP-Growth Step 4
 - For each frequent 1-itemset, grow each itemset while recursively building its conditional FP-trees

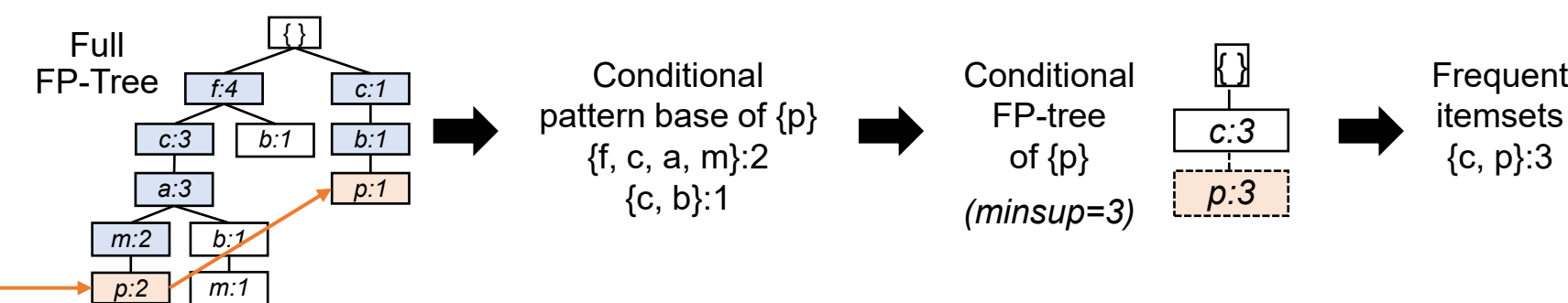


- Frequent k-itemsets

1-itemset	Conditional pattern bases	Conditional FP-tree	Frequent k-itemsets (k>1)
f:4			
c:4			
a:3			
b:3			
m:3			
p:3			

Pattern Mining > Methodologies > FP-Growth

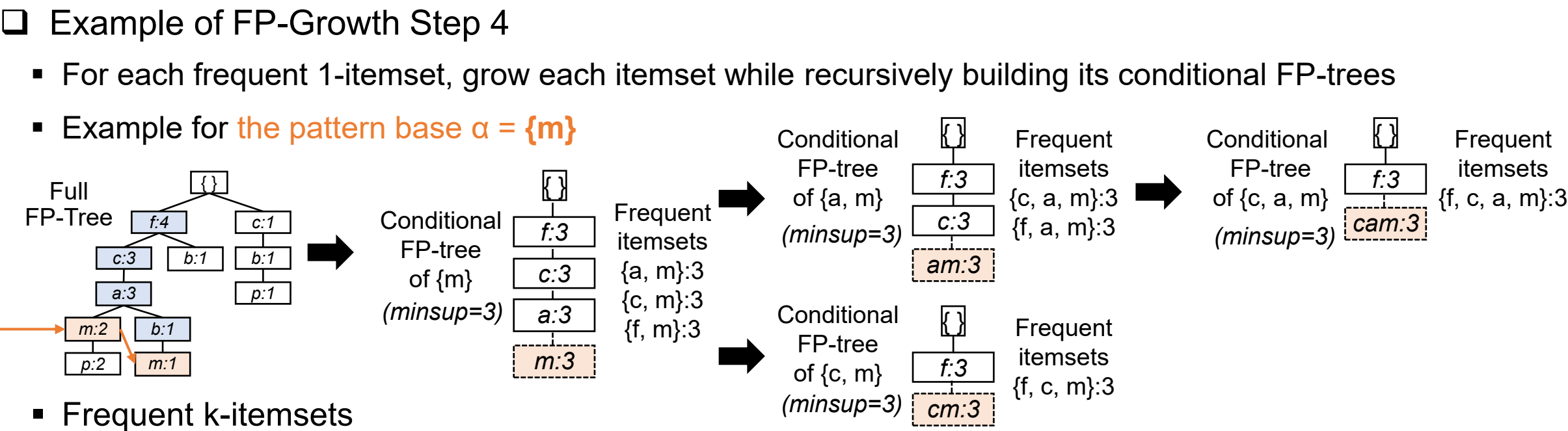
- ❑ Example of FP-Growth Step 4
 - For each frequent 1-itemset, grow each itemset while recursively building its conditional FP-trees
 - Example for the pattern base $\alpha = \{p\}$



- Frequent k-itemsets

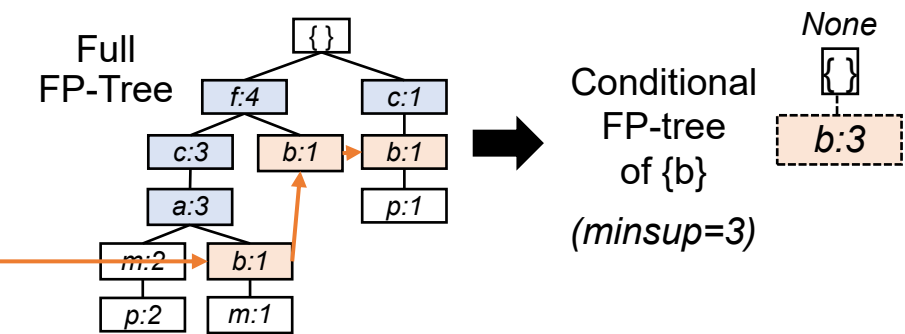
1-itemset	Conditional pattern bases	Conditional FP-tree	Frequent k-itemsets (k>1)
f:4			
c:4			
a:3			
b:3			
m:3			
p:3	{f, c, a, m}:2, {c, b}:1	c:3	{c, p}: 3

Pattern Mining > Methodologies > FP-Growth



Pattern Mining > Methodologies > FP-Growth

- ❑ Example of FP-Growth Step 4
 - For each frequent 1-itemset, grow each itemset while recursively building its conditional FP-trees
 - Example for the pattern base $\alpha = \{b\}$

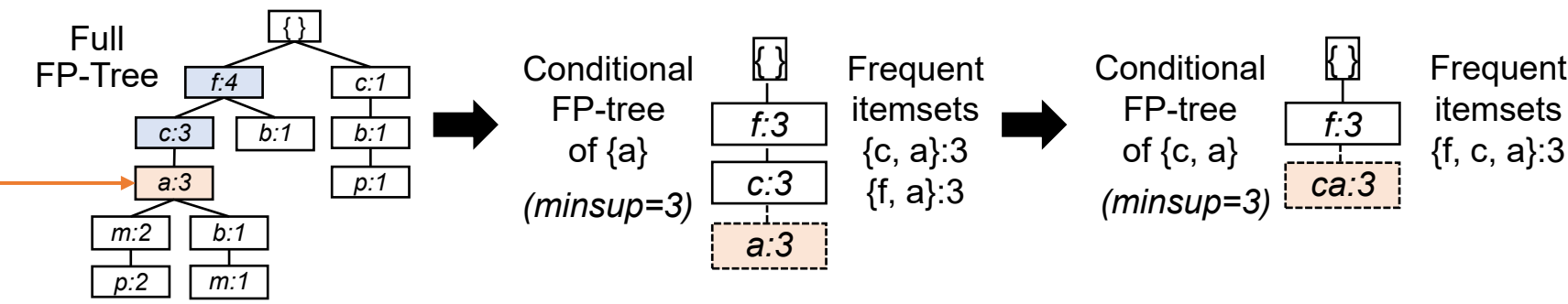


- Frequent k-itemsets

1-itemset	Conditional pattern bases	Conditional FP-tree	Frequent k-itemsets (k>1)
f:4			
c:4			
a:3			
b:3	{f, c, a}:1, {f}:1, {c}:1	None	None
m:3	{f, c, a}:2, {f, c, a, b}:1	f:3 – c:3 – a:3	{f, m}:3, {c, m}:3, {a, m}: 3, {f, c, m}:3, {f, a, m}:3, {c, a, m}:3,{f, c, a, m}:3
p:3	{f, c, a, m}:2, {c, b}:1	c:3	{c, p}: 3

Pattern Mining > Methodologies > FP-Growth

- ❑ Example of FP-Growth Step 4
 - For each frequent 1-itemset, grow each itemset while recursively building its conditional FP-trees
 - Example for the pattern base $\alpha = \{a\}$

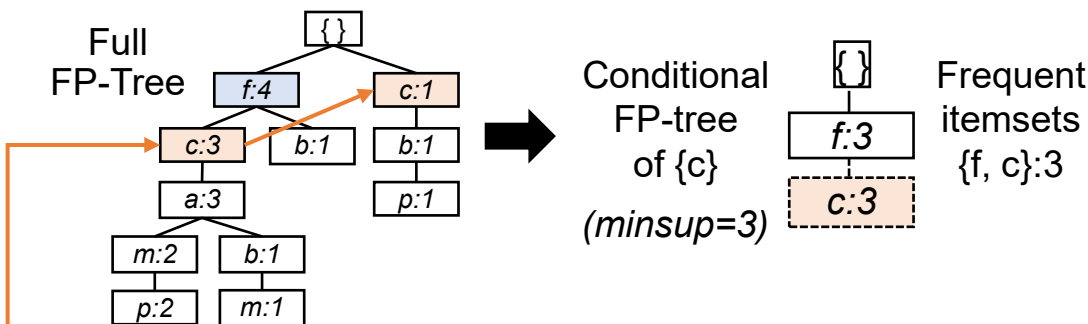


- Frequent k-itemsets

1-itemset	Conditional pattern bases	Conditional FP-tree	Frequent k-itemsets (k>1)
f:4			
c:4			
a:3	{f, c}:3	f:3 – c:3	{f, a}:3, {c, a}:3, {f, c, a}:3
b:3	{f, c, a}:1, {f}:1, {c}:1	None	None
m:3	{f, c, a}:2, {f, c, a, b}:1	f:3 – c:3 – a:3	{f, m}:3, {c, m}:3, {a, m}: 3, {f, c, m}:3, {f, a, m}:3, {c, a, m}:3,{f, c, a, m}:3
p:3	{f, c, a, m}:2, {c, b}:1	c:3	{c, p}: 3

Pattern Mining > Methodologies > FP-Growth

- ❑ Example of FP-Growth Step 4
 - For each frequent 1-itemset, grow each itemset while recursively building its conditional FP-trees
 - Example for the pattern base $\alpha = \{c\}$

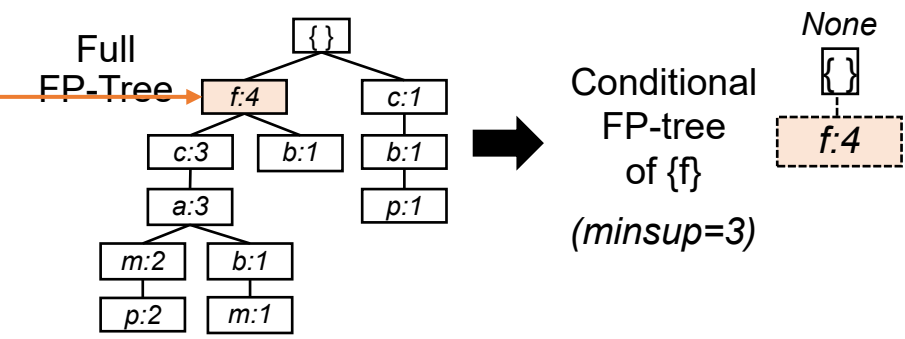


- Frequent k-itemsets

1-itemset	Conditional pattern bases	Conditional FP-tree	Frequent k-itemsets (k>1)
f:4			
c:4	{f}:3	f:3	{f, c}:3
a:3	{f, c}:3	f:3 – c:3	{f, a}:3, {c, a}:3, {f, c, a}:3
b:3	{f, c, a}:1, {f}:1, {c}:1	None	None
m:3	{f, c, a}:2, {f, c, a, b}:1	f:3 – c:3 – a:3	{f, m}:3, {c, m}:3, {a, m}: 3, {f, c, m}:3, {f, a, m}:3, {c, a, m}:3,{f, c, a, m}:3
p:3	{f, c, a, m}:2, {c, b}:1	c:3	{c, p}: 3

Pattern Mining > Methodologies > FP-Growth

- ❑ Example of FP-Growth Step 4
 - For each frequent 1-itemset, grow each itemset while recursively building its conditional FP-trees
 - Example for the pattern base $\alpha = \{f\}$

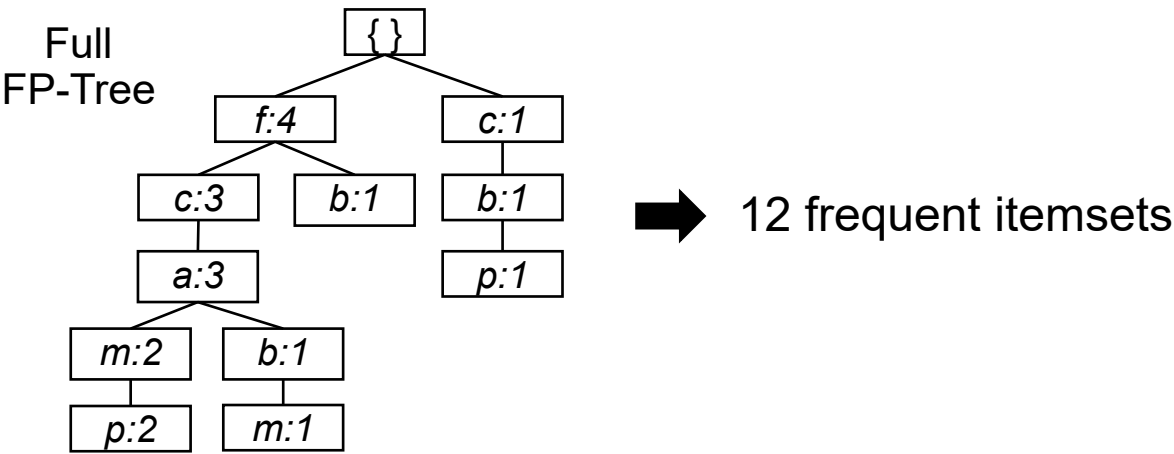


- Frequent k-itemsets

1-itemset	Conditional pattern bases	Conditional FP-tree	Frequent k-itemsets (k>1)
f:4	None	None	None
c:4	{f}:3	f:3	{f, c}:3
a:3	{f, c}:3	f:3 – c:3	{f, a}:3, {c, a}:3, {f, c, a}:3
b:3	{f, c, a}:1, {f}:1, {c}:1	None	None
m:3	{f, c, a}:2, {f, c, a, b}:1	f:3 – c:3 – a:3	{f, m}:3, {c, m}:3, {a, m}: 3, {f, c, m}:3, {f, a, m}:3, {c, a, m}:3,{f, c, a, m}:3
p:3	{f, c, a, m}:2, {c, b}:1	c:3	{c, p}: 3

Pattern Mining > Methodologies > FP-Growth

- ❑ Example of FP-Growth Step 4
 - For each frequent 1-itemset, grow each itemset while recursively building its conditional FP-trees



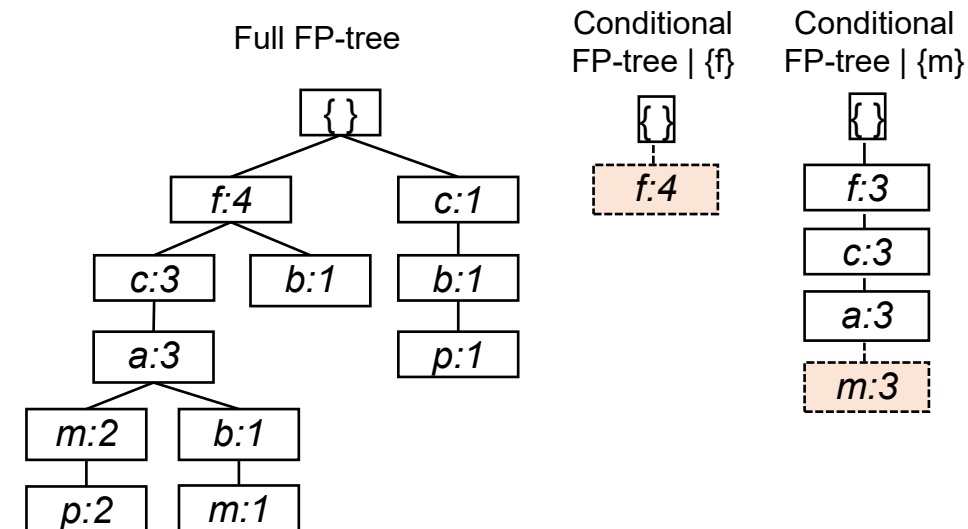
- Final frequent k-itemsets

1-itemset	Conditional pattern bases	Conditional FP-tree	Frequent k-itemsets (k>1)
f:4	None	None	None
c:4	{f}:3	f:3	{f, c}:3
a:3	{f, c}:3	f:3 – c:3	{f, a}:3, {c, a}:3, {f, c, a}:3
b:3	{f, c, a}:1, {f}:1, {c}:1	None	None
m:3	{f, c, a}:2, {f, c, a, b}:1	f:3 – c:3 – a:3	{f, m}:3, {c, m}:3, {a, m}: 3, {f, c, m}:3, {f, a, m}:3, {c, a, m}:3,{f, c, a, m}:3
p:3	{f, c, a, m}:2, {c, b}:1	c:3	{c, p}: 3

Pattern Mining > Methodologies > FP-Growth

□ Recap of FP-Growth

- Find frequent 1-itemsets (1st scan)
- Construct a customized data structure, FP-tree (2nd scan)
- For each frequent 1-itemset,
 - Partition the TDB based on them (i.e., conditional pattern base)
 - Recursively exploit FP-tree (i.e., conditional FP-tree) to grow frequent patterns from each partition
 - Stop conditions
 - The resulting FP-tree is empty (e.g., FP-tree | {f}))
 - No further pattern base exists
 - The resulting FP-tree contains only one path (e.g., FP-tree | {m})
 - All the combinations of its sub-paths become frequent patterns



Pattern Mining > Methodologies > FP-Growth

□ Pseudo-code of FP-Growth

- Note that the support thresholding on the f-list is omitted here

minsup

```
Procedure: FPGrowth(DB,  $\xi$ )
Define and clear F-List : F[];
foreach Transaction Ti in DB do
    foreach Item aj in Ti do
        F[ai] ++;
    end
end
Sort F[];
Define and clear the root of FP-tree : r;
foreach Transaction Ti in DB do
    Make Ti ordered according to F;
    Call ConstructTree(Ti, r);
end
foreach item ai in I All items do
    Call Growth(r, ai,  $\xi$ );
end
```

(tree, base item, minsup)

```
Procedure: Growth(r, a,  $\xi$ )
if r contains a single path Z then
    foreach combination(denoted as  $\gamma$ ) of the nodes in
        Z do
        Generate pattern  $\beta = \gamma \cup a$  with support =
            minimum support of nodes in  $\gamma$ ;
        if  $\beta.support > \xi$  then
            Call Output( $\beta$ );
        end
    end
else
    foreach bi in r do
        Generate pattern  $\beta = b_i \cup a$  with support =
            bi.support;
        if  $\beta.support > \xi$  then
            Call Output( $\beta$ );
        end
        Construct  $\beta$ 's conditional database ;
        Construct  $\beta$ 's conditional FP-tree Tree $\beta$ ;
        if Tree $\beta$   $\neq \phi$  then
            Call Growth(Tree $\beta$ ,  $\beta$ ,  $\xi$ );
        end
    end
end
```

Outline

- ❑ Basic concepts of pattern mining
- ❑ **Pattern mining methodologies**
 - Association rule discovery
 - Apriori
 - FP-Growth
 - **Rule generation**
- ❑ Pattern mining evaluation

Pattern Mining > Methodologies > Rule Generation

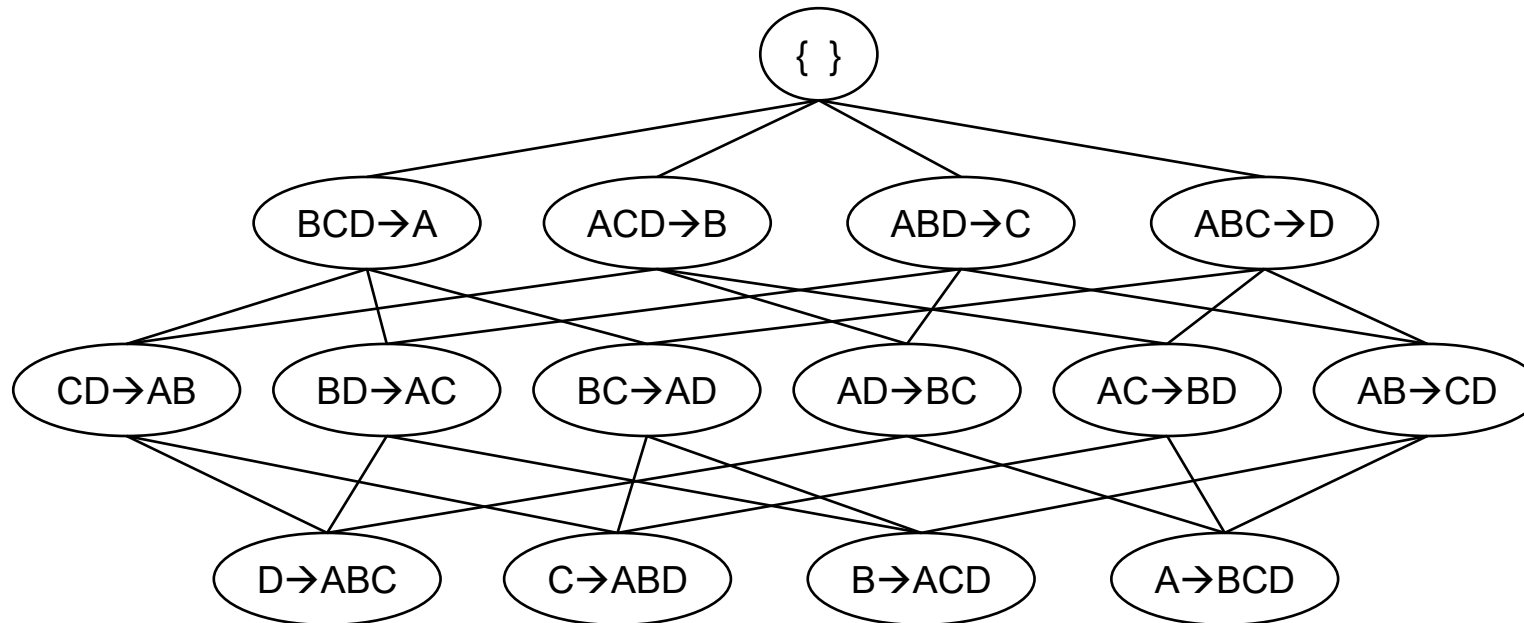
□ Given a frequent itemset X ,

- Possible association rules

- All non-empty subsets $Y \subset X$ such that $X - Y \rightarrow Y$ satisfies the minimum confidence threshold
- If $|X| = k$, then there are $2^k - 2$ candidate association rules (except for $X \rightarrow \emptyset$ and $\emptyset \rightarrow X$)

- Example

- If $X = \{A, B, C, D\}$ is a frequent itemset, candidate rules can be:



■ Pattern Mining > Methodologies > Rule Generation

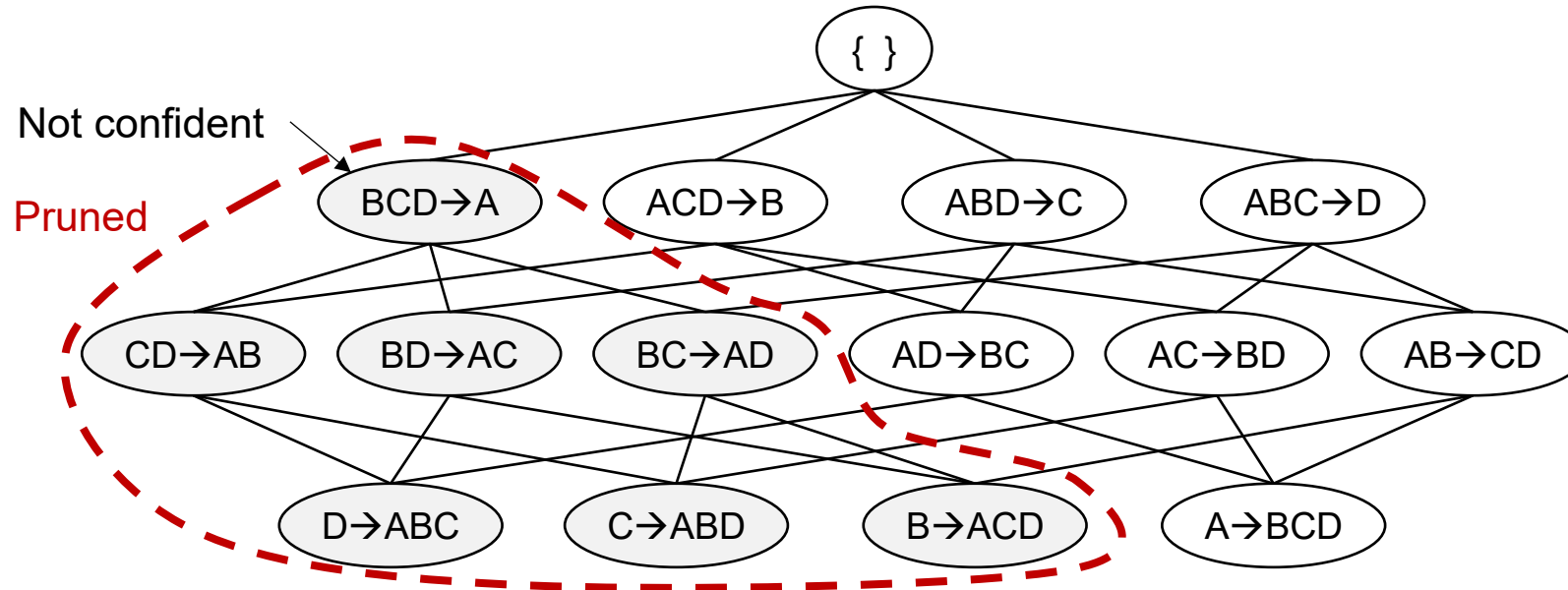
□ How to efficiently generate rules from frequent itemsets?

- Unlike support, confidence does not have an anti-monotone property
- Example $X = \{A, B, D\} \subseteq Y = \{A, B, C, D\}$
 - Support
 - $s(\{A, B, D\}) \geq s(\{A, B, C, D\})$
 - Confidence
 - Frequent 4-itemset $\{A, B, C, D\}$: $c(ABC \rightarrow D) = s(\{A, B, C, D\})/s(\{A, B, C\})$
 - Frequent 3-itemset $\{A, B, D\}$: $c(AB \rightarrow D) = s(\{A, B, D\})/s(\{A, B\})$
 - $c(ABC \rightarrow D) ?? c(AB \rightarrow D)$

Pattern Mining > Methodologies > Rule Generation

□ How to efficiently generate rules from frequent itemsets?

- Confidence does not have an anti-monotone property
 - e.g., $c(ABC \rightarrow D)$ can be larger or smaller than $c(AB \rightarrow D)$
- Confidence of rules generated from the same itemset has an anti-monotone property
 - Confidence is anti-monotone w.r.t. the number of items on the right-hand-side of the rule
 - e.g., $X = \{A,B,C,D\}$: $c(BCD \rightarrow \mathbf{A}) \geq c(CD \rightarrow \mathbf{AB}) \geq c(D \rightarrow \mathbf{ABC})$



■ Outline

- ❑ Basic concepts of pattern mining
- ❑ Pattern mining methodologies
- ❑ **Pattern mining evaluation**
 - Motivation
 - Lift

■ Pattern Mining > Evaluation > Motivation

□ *Are support and confidence sufficient?*

■ Example

- TDB with minsup = 30% and minconf = 60%
 - Total 10,000 transactions
 - 6,000 transactions include “game”
 - 7,500 transactions include “video”
 - 4,000 transactions include both “game” and “video”
- A strong rule discovered: {“game”} $\rightarrow$ {“video”}
 - Support = $4,000/10,000 = 40\% > \text{minsup}$
 - Confidence = $4,000/6,000 = 66\% > \text{minconf}$
- Is it interesting?
 - It can be misleading because the probability of purchasing “video” is $7,500/10,000 = 75\% > 66\%$

■ Pattern Mining > Evaluation > Lift

□ Correlation measure: Lift

- The degree to which the occurrence of one “lifts” the occurrence of the other

$$lift(A, B) = \frac{c(A \rightarrow B)}{s(B)} = \frac{s(A \cup B)}{s(A) \times s(B)} = \frac{P(A \cup B)}{P(A) \times P(B)}$$

- $P(A \cup B) = P(A) \times P(B) \rightarrow \text{Lift} = 1$: A and B are independent
- $P(A \cup B) > P(A) \times P(B) \rightarrow \text{Lift} > 1$: A and B are positively correlated
- $P(A \cup B) < P(A) \times P(B) \rightarrow \text{Lift} < 1$: A and B are negatively correlated
- Example
 - $\text{Lift}(\text{game}, \text{video}) = 0.4 / (0.6 \times 0.75) = 0.89 < 1$
 - The sale of game is said to increase the likelihood of the sale of video by a factor of lift = 0.89
 - Negatively correlated $\rightarrow$ the sale of game does not help the sale of video
- It is more informative if we have the lift score in addition to the support and confidence
 - “A $\rightarrow$ B” [support, confidence, lift]

Tutorial References

□ Examples

- [Association rule mining example \(FP-growth\)](#)
- [Association rule mining example \(Apriori\)](#)
- [Market-basket analysis example \(Apriori\)](#)

□ Library

- [Mlxtend library – FP-Growth](#)
- [Mlxtend library – Apriori](#)

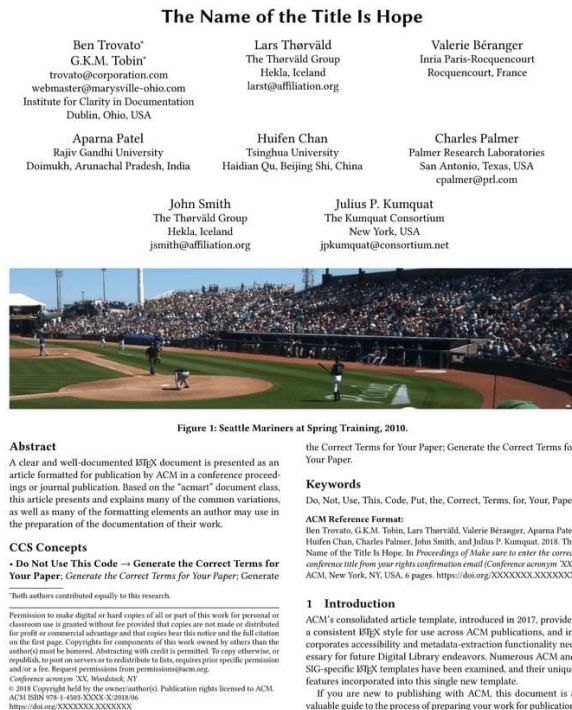
□ Implementations

- [FP-growth implementation](#)
- [Apriori implementation](#)

Next: Tutorial 1

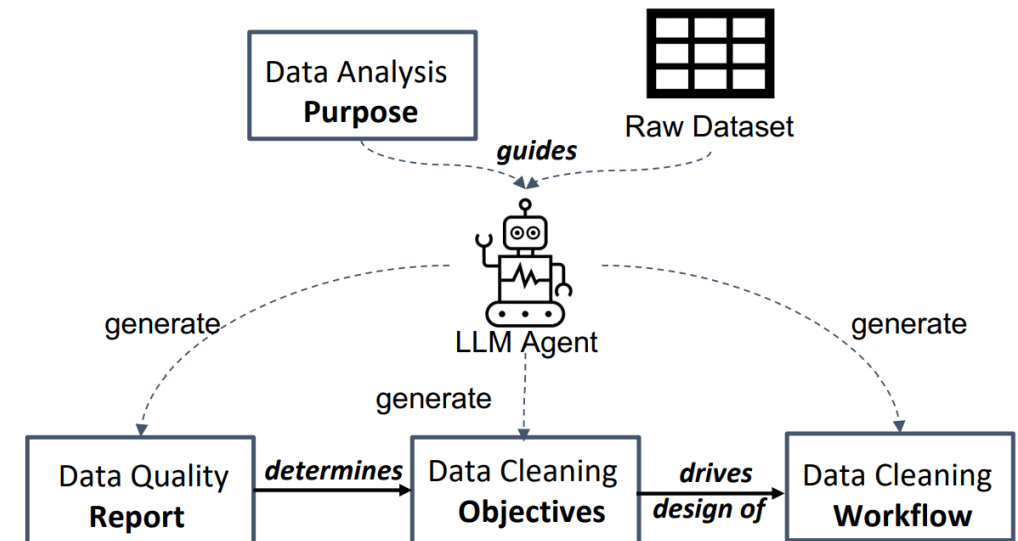
□ How to write a paper with Latex

- <http://www.overleaf.com>



□ Preprocessing with agents

- <http://colab.google.com>
- [AutoDCWorkflow: LLM-based Data Cleaning Workflow Auto-Generation and Benchmark \(EMNLP '25\)](#)



[Answer] Pattern Mining > Methodologies > Apriori

❑ Quiz: how many frequent k-itemsets ($k > 1$) are there? **Answer: 7**

TDB

TID	Items bought together
1	Beer, Coffee, Nuts, Diaper, Milk
2	Beer, Coffee, Diaper
3	Beer, Nuts, Diaper, Eggs
4	Nuts, Diaper, Milk
5	Beer, Nuts, Diaper, Milk

minsup = 60% → 3

C ₁			F ₁			C ₂			F ₂			C ₃			F ₃			C ₄			F ₄		
Itemset	Itemset	Sup	Itemset	Sup		Itemset	Itemset	Sup	Itemset	Sup		Itemset	Itemset	Sup	Itemset	Sup		Itemset	Itemset	Sup	Itemset	Sup	
{Beer}	{B}	4				B,N	B,N	3	B,N,D	3		B,N,D	B,N,D	3	B,N,D,M								
{Nuts}	{N}	4				B,D	B,D	4	B,N,M	2		N,D,M	N,D,M	3									
{Diaper}	{D}	5				B,M	B,M	2	N,D	4													
{Coffee}	{C}	2				N,D	N,D	4	N,M	3													
{Eggs}	{E}	4				N,M	N,M	3	D,M	3													
{Milk}	{M}	3				D,M	D,M	3															

Cookie Slides - Frequent Pattern in Practice

- Keyword recommendation at Naver Line Japan App (prototype)
 - Effective search keyword suggestion leads to more consumption of relevant content

Rolling search keywords suggestion

Dog
BTS Coldplay
Twice New Single
...



Cookie Slides - Frequent Pattern in Practice

- Keyword recommendation at Naver Line Japan App (prototype)
 - Not enough user query logs? → content-based keyword pattern extraction

Content (e.g., news articles)

컨텐츠 A

설 틈이 없어요 쭉쭉쭉~방탄소년단, 유엔총회→콜드플레이 콜라보(종합)[...]
어나더 월드 클래스 **BTS**이기에 가능한 일이다. 방탄소년단은 24일 오전 문재인 대통령의 특별 사절(특사)... 이날 오후 콜드플레이와 콜라보레이션 한 곡 'My Universe'가 공개되자 공식 ...



컨텐츠 B

BTS·콜드플레이, 콜라보 음원 발매..."My Universe, 한국어·영어 가사"
방탄소년단과 영국 밴드 콜드플레이가 역대급 콜라보레이션을 공개했다. 방탄소년단과 콜드플레이가 24일 오후 1시 콜라보레이션 싱글 '마이 유니버스'(My Universe)를 전 세계에 발표...



컨텐츠 C

BTS·콜드플레이 콜라보 곡 '마이 유니버스' 오늘 공개
그룹 방탄소년단(**BTS**)이 세계적 브릿팝 밴드 콜드플레이(Coldplay)와의 컬래버레이션 곡 '마이 유니버스'를... 부분을 **BTS** 멤버들이 불렀습니다. **BTS**는 한국어와 영어 가사를 통해 우주...



컨텐츠 D

베일 벗은 콜드플레이·BTS의 콜라보 '마이 유니버스'... 한국어 가사도 귀...
영국의 세계적 록 밴드 콜드플레이(Coldplay)가 현재 최고 인기를 달리고 있는 그룹 방탄소년단(**BTS**)과 함께... 콜드플레이 멤버들과 **BTS**의 RM, 슈가, 제이홉이 작사·작곡자에 이름을 ...



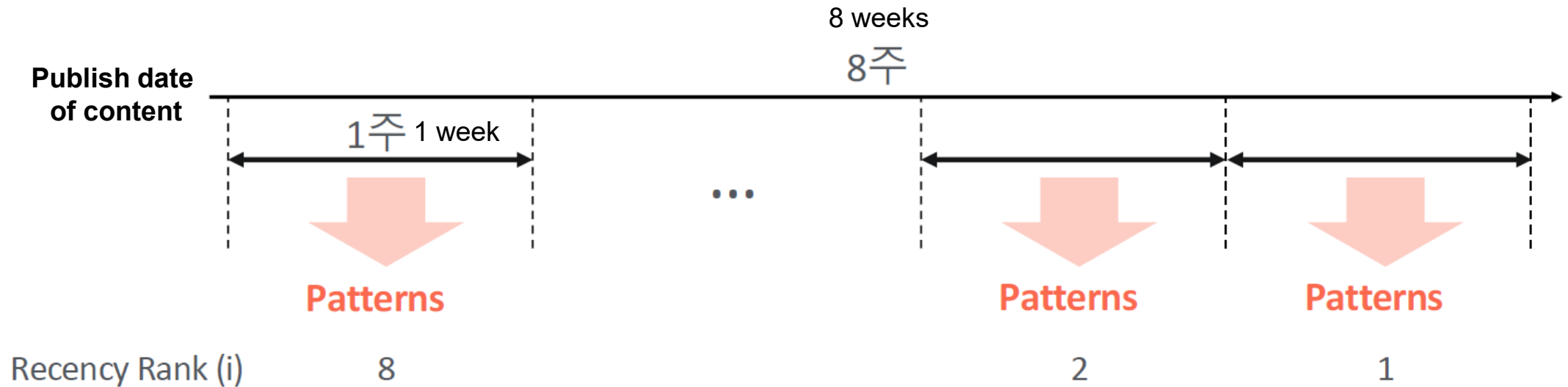
Sequential Patterns

...
Coldplay
(BTS, 콜드플레이)
Collabo
(BTS, 콜라보)
(BTS, My Universe)
Coldplay Collabo
(BTS, 콜드플레이, 콜라보)
Coldplay
(콜드플레이, My Universe)
...



Cookie Slides - Frequent Pattern in Practice

- Keyword recommendation at Naver Line Japan App (prototype)
 - Extension 1: considering time-aware support



Temporally decaying weighted support

각 패턴에 대한 점수 계산

$$\sum_{i=1}^8 \frac{\text{support}}{\log_2(i+1)}$$

여러 구간에서 뵈힐수록, More likely to be found in multiple periods,
각 구간에서의 support (출현 빈도) 값이 클수록, More frequently found in each period,
최근에 뵈힐수록 값이 커짐 More recently found,
→ Higher support score

Cookie Slides - Frequent Pattern in Practice

- ❑ Keyword recommendation at Naver Line Japan App (prototype)
 - Extension 2: considering window size of patterns constraint

Content title				
English Single Music Video 2 nd Teaser Out!				
「TWICE」 영어 싱글 「The Feels」 뮤직비디오 티저 2탄 공개!				
← Window size = 4 (e.g., 4 keywords) → ← Window size = 7 (e.g., 7 keywords) →				
Example patterns	Window Size: 5	Window Size: 6	Window Size: 7	Not considered Window 고려 안함
	1 (twice, 일본) Japan	(twice, itzy)	(twice, itzy)	(twice, 공개) Out
	2 (twice, 정연) JeongYeon	(twice, 정연)	(twice, 정연)	(twice, itzy)
	3 (twice, 나연) NaYeon	(twice, 나연)	(twice, 나연)	(twice, 정연)
	4 (twice, 일본, 데뷔) Japan, Debut	(twice, 일본, 데뷔)	(twice, 일본, 데뷔)	(twice, 나연)
	5 (twice, 동생뺄) Younger age	(twice, 동생뺄, itzy)	(twice, 동생뺄, itzy)	(twice, 멤버)
	6 (twice, itzy, 일본) Japan	(twice, 모모)	(twice, 모모)	(twice, 일본, 데뷔)
	7 (twice, 모모) MoMo	(twice, perfectworld)	(twice, perfectworld)	(twice, 동생뺄, itzy)
	8 (twice, perfectworld)	(twice, 썬위)	(twice, 썬위)	(twice, 모모)
	9 (twice, 썬위) Tzuyu	(twice, 일본)	(twice, 일본)	(twice, 주목)
	10 (twice, 워너뮤직재팬, 일본)	(twice, 사나)	(twice, 공개) Out	(twice, thefeels)
WarnerMusicJapan, Japan				

Cookie Slides - Frequent Pattern in Practice

□ Keyword recommendation at Naver Line Japan App (prototype)

- Application to personalized keyword recommendation

